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INTRODUCTION

This procedure provides guideline information for the installation of an MS6001B Gas Turbine-Generator unit. These guidelines apply to a typical centerline installation.

The installation of many commonly purchased off-base modules and skids (listed in Table 1) are addressed in separate Field Engineer's Manual articles.

Assumptions

- The user of this procedure is an experienced field engineer.
- The user is familiar with the general configurations and functions of the module packages and the Model List Item (MLI) structure.
- The site preparations listed below have been completed by others, and the site is "foundation ready" in accordance with the following:

Site Preparations (Completed by Others)

- Site survey
- Geotechnical investigations
- Rough Excavation and grading
- Installation of main and auxiliary foundations (including all required embedments and grout pockets)
- Drain collection headers and tanks
- Temporary electrical power, water, grounding, etc.

Responsibilities

The GE on-site representative provides technical direction and related advice. The field engineer should always promote good working practices and safe working conditions. The field engineer's responsibility is limited to the installation and field alignment of equipment and materials supplied by Gas Turbine-Generator (GT-G). This includes providing the customer with technical counsel and advice. The field engineer should comment on the work and materials supplied by others if aware of an unacceptable or potentially detrimental situation or practice.

Much of the following information is general in nature and does not cover all details or provide for every contingency, and should be used in conjunction with other documents referenced here. Technical requirements supplied by GT-G take precedence over these installation guidelines.

INITIAL ACTIVITIES

The following administrative activities must be completed by the technical director/advisor before setting the main equipment packages:

- Pre-Installation Meeting
- Site Office Set-up
- Accumulation of Installation Drawings (see EM 5016 – Installation Documents)
- Review of Construction Tools Required on Site
- Review of Precision Tools Required for Alignment
- Fabrication/Procurement of Special Tools
- Material Inventory, Inspection and Storage

FOUNDATION INTERFACE ACTIVITIES

Although GE Energy Services may not have responsibility for foundation installation, it is critical that it be thoroughly inspected prior to setting equipment. This inspection is required to verify that the foundation has been properly formed with anchor bolts, grout pockets, restraining blocks, jacking posts, conduit stub-ups and cable pit/vault properly located as required for installation of the turbine-generator unit. Additionally, the inspection must verify that centerline references are established for use during initial setting of equipment. For information on the specific inspections required refer to EM 5028 – Inspection of Site Interfaces.

Another area that is critical to short term and long term successful operation of a gas turbine-generator unit is setting and grouting the sub-sole plates. Since these plates and associated grout support the turbine-generator and maintain the relative alignment throughout the life of the unit, it is vital that the work be performed correctly and completely. Refer to EM 5036 – Setting Sub-Sole Plates and Shims. (See Figs. 1, 2, 3, and 4).

Other applicable documents provided by GT-G include:

MLI 0306 – Outline, Mech. GTRB & Load
MLI 0323 – Foundation Interface Outline
MLI 1603 – Foundation Bolting Arrangement

TABLE 1 — UNIT CONFIGURATION
MS6001B "Split Base" units may include the following equipment packages

MLI	PACKAGES	COMMENTS
0406	Turbine Compartment	Now separate from the accessory
0975	Accessory Compartment	New smaller sized for MS 61B
0991	Gas Fuel Module	Requires off base hydraulic piping
1101	Control Compartment	Usually includes MCC's; however a separate PEECC may be included
1233	Exhaust Frame Blowers	Two per turbine
A012	Load Gear	Philadelphia Gear Co.
A013	Generator, 6A3 or 6A6	Open ventilated or TEWAC (Totally Enclosed Water to Air Cooling)
A025	Auxiliary Transformer	Transformer and switch gear lineup
A035	Water Injection Skid	Single-stage or two-stage
A036	Gen. Auxiliary Cmpt.	GE or PACS Industries (GAC)
A040	Turb. Inlet Air Filter	Donaldson, Pneumafil or cmpt. Farr
A041	Turbine Inlet Air Duct	Including support steel
A042	Exhaust Duct And Stack	Straight up or side mounted
A045	Fuel Oil Forwarding	Usually located near fuel module, strage tanks
A098	Lube Oil Vapor Demister	Standard off-base
A068	Fire Protection Skid	CO2 bottles or tank
A076	Cooling Water Module	Normally a two-fan configuration. May not be included on industrial/combined cycle units
A087	Starting Transformer	Transformer and switchgear lineup (rarely used. Most units are diesel start)
A130	Process "Pulse" Air	Back pressure cleaning of air skid filters



Figure 1. Shooting Foundation Embedment Plates and Sub-Sole Plate Elevation

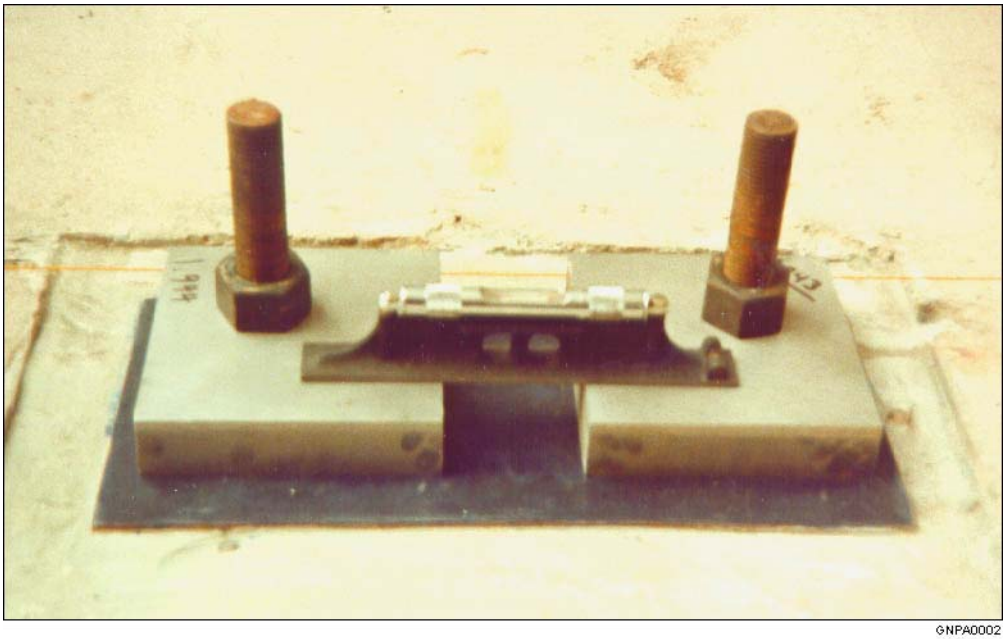


Figure 2. Checking Sub-Sole Plates Flat and Level (w/Model 98 Master Level)

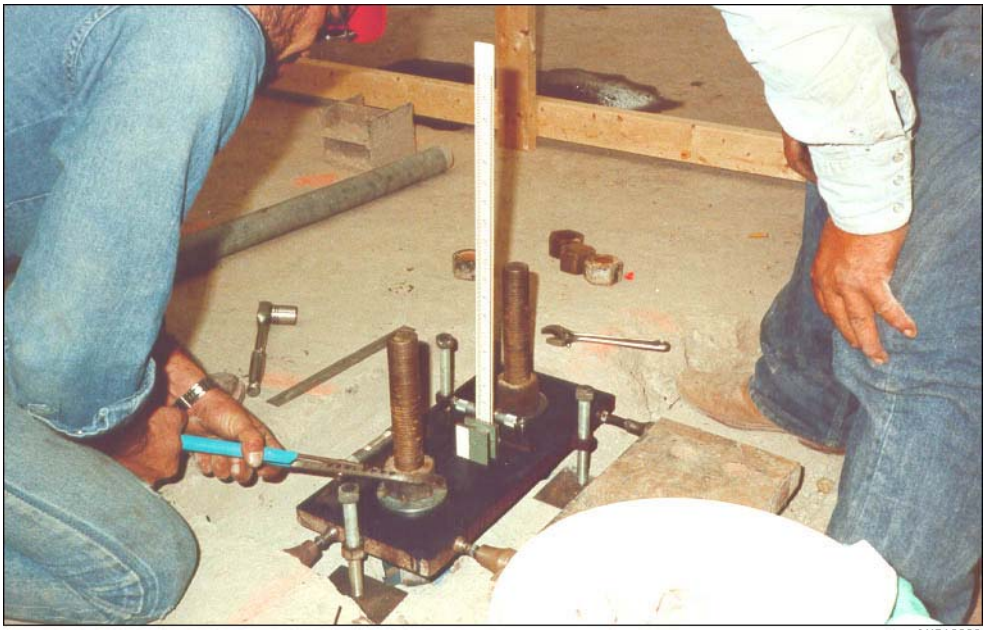


Figure 3. Adjusting Anchor Bolt Nut During Setting of Sub-Sole Plate with Precision Tool



Figure 4. Pouring Grout into Foundation Grout Pocket Around Sub-sole Plate

RECEIVING INSPECTION – GENERAL

On receipt of the turbine, generator and accessory compartment, check the factory installed accelerometers ("hump meters") to determine whether the generator may have been humped during shipment. These meters consist of two springs and four ball bearings in a plastic cover. If the springs are sprung, and the balls are no longer supported by the springs, the generator experienced a rough ride and damage may have occurred. Report this to Installation Engineering.

SETTING MAIN PACKAGES

The sequence in which the main packages are set is often a matter of the personal preference of the field engineer, or determined by which piece of equipment arrives on site first. This guideline procedure has chosen a sequence which has proven satisfactory during past installations. This procedure describes a sequence in which both the turbine and the generator are set before the load gear is set. This procedure allows setting the generator with the rotor shipping lock in place. Prior to setting the load gear, the generator rotor locking device will be removed and the rotor (field) moved aft to provide sufficient clearance for setting the load gear.

The following activities are common to setting each base:

- Clean the concrete foundation of debris and dirt.
- Clean all sub-sole plates and the top surface of centerline restraining blocks.
- Protect the anchor bolt threads by removing the washers and nuts and sliding lengths of plastic pipe over the bolt threads. (See Fig. 5).
- After each base is set, remove the plastic pipe, assemble the anchor bolt washers and nuts, and torque the nuts. (See Table 2).
- Use a hammer to sound check for lightly loaded shim packs. (Refer to EM 1081 – Grouting Foundation Plates).
- Use a feeler gauge to determine the percent contact and/or the number of makeup shims needed. Adjust the shims as required (Refer to EM 5036 – Setting Sub-sole plates, for generator shim pack preparation).

NOTE: The 6A3 or 6A6 generator typically requires an additional 0.800 inch above the Machine Base Line of the gas turbine and load gear. (See MLI 0323 – Foundation Interface Outline, for specific requirements).



Figure 5. Protective Sleeves Over Anchor Bolts

Setting the Generator Base

1. Secure lifting cables around the four trunnions located in the generator base structure, and lift the generator into position over the foundation.

NOTE: Another common setting method is to slide the generator on rails or with rollers, and to raise or lower the unit using jacks and cribbing.

2. While on blocking clean, debur and hone the generator base sole plates.
3. Lower the generator onto the foundation sub-sole plate shim packs, keeping the anchor bolts centered in the holes in the base. (See Fig. 6).

   Refer to MLI 1603 – Foundation Bolting arrangement, MLI 0306 – Outline Mechanical Gas turbine and Load, and MLI 0323 – Outline, Foundation Interface for shim pack thickness and calculations.
4. Position the centerline keyways at the ends of the base on the centerline restraining blocks in the foundation. (The locating centerline guide blocks and "U-shaped" keys will be positioned after the unit is aligned).
5. Ensure that the base is moved as a rigid body to minimize axial flexing. Jacking should be done in close proximity to the base lifting trunions.
6. Remove the studs, and the rotor shipping lock (painted yellow). The lock is used to secure the turbine end of the generator rotor to the pedestal during shipment (see Fig. 7).



Figure 6. Lowering Generator on Foundation

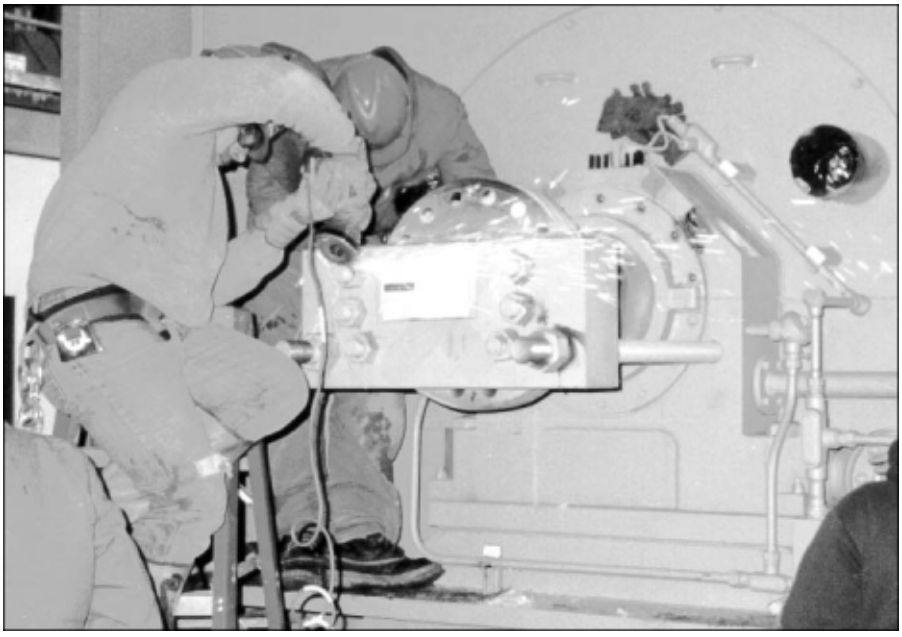


Figure 7. Removing Rotor Shipping Lock

7. Remove the vibration measurement proximity probes from the turbine-end pedestal (if supplied), to prevent damage when moving the rotor. Sufficient clearance (approximately one inch total), is designed into the generator to allow the rotor to be moved axially toward the collector end for coupling disengagement.

8. Move the rotor axially toward the collector end of the generator approximately 0.750 inch from the normal cold set position. (See Fig. 8). This allows the coupling flange faces sufficient clearance during setting of the load gear.

CAUTION: Do not allow contact between rotor and stator components when moving the generator field (rotor). Refer to the specific generator mechanical outline for the actual clearance available. (The 6A3 Generator is typically installed with the MS 6001B Gas Turbine).

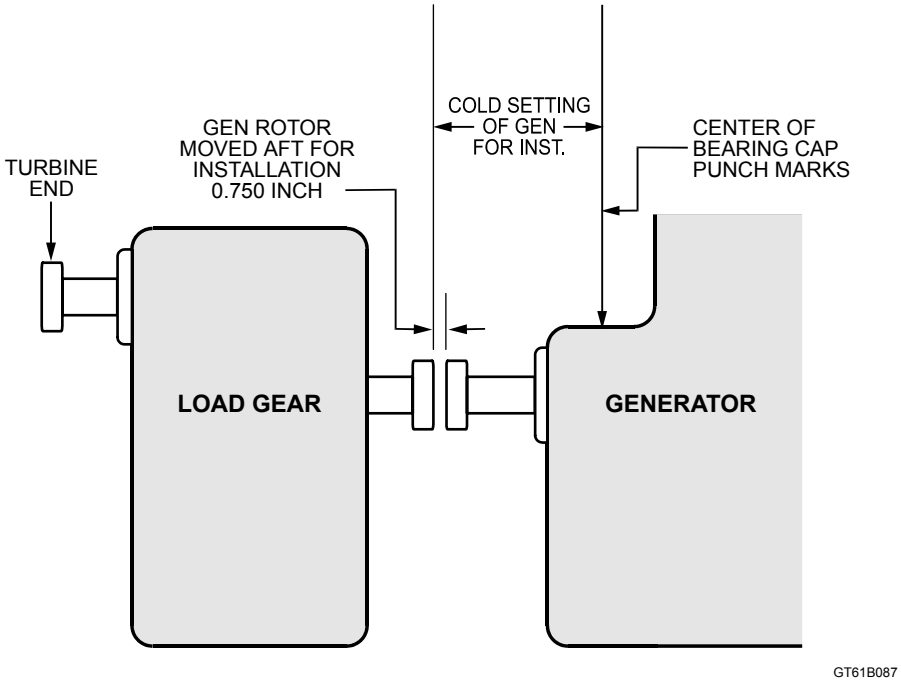
9. Use a micrometer to measure the generator hub rabbit fits. Take four measurements every 45 degrees and average to find each dimension.
(See Fig. 9. Rabbet Interference Fit Measurement).

Table 2
Foundation Bolt Torque Values (Reference MLI 1603)

Bolt Dia. (Inches)	0.625	0.750	0.875	1.000	1.250	1.500	1.750	2.000
Torque (Ft-Lb.	67	100	175	260	565	1000	1650	2500
Torque (M-KG)	9.3	13.8	24.2	36.0	78.1	138.3	228.3	345.8

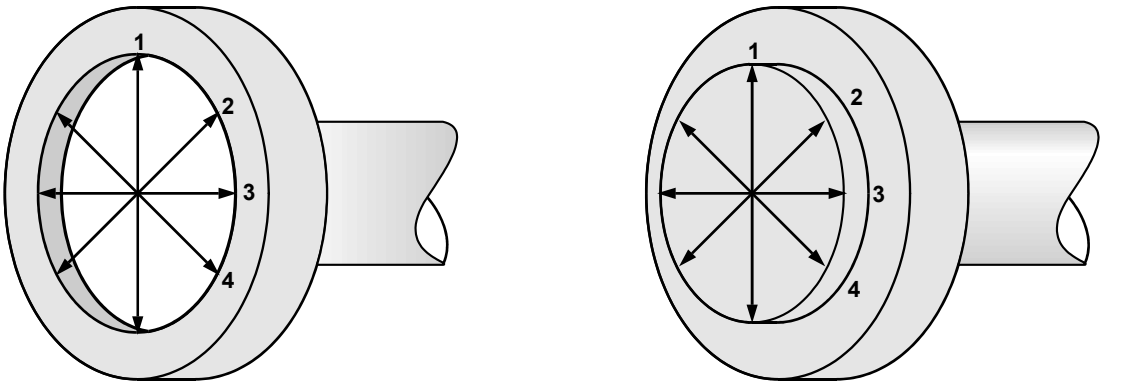
Note: These torque values are based on two thirds of yield strength for the ASTM A325 or A307 anchor bolt material suggested by GT, or a stress of approximately 7000 psi.

Caution: Move the base as a rigid body to minimize axial flexing. Jacking should be done in close proximity to the base lifting trunnions.



Note: See Generator Mechanical Outline drawing for specific dimensions.

Figure 8. 6A3 Generator Filed Position Data (Typical)



Female Rabbet Fit Measurement. **Male Rabbet Fit Measurement.**

Figure 9. Rabbet Interference Fit Measurement

10. When the generator is properly positioned to the foundation centerlines, install the anchor bolt nuts and washers, and torque to the maximum torque recommended (See Table 2). Typically, the anchor bolt washers are at least one (1) inch thick. Check the torque periodically. Once final alignment has been achieved, the anchor bolt nuts are usually loosened to provide a 0.010 inch clearance between the washer and the nut, per the Generator Mechanical Outline Drawing.

Note: If the generator is stored or will be on the foundation for a week or more, the generator heaters should be energized. However, the heaters must be deenergized for a minimum of 24 hours prior to starting alignment to the load gear.

Setting the Accessory Base

1. Immediately before setting the accessory base, verify that the mechanical centerline has been scribed across the forward and aft restraining block embedments and is easily visible.
2. Remove the 3/16 inch thick steel cover, tack welded on the accessory gear end.
3. Place the forward accessory support guide and the aft accessory support plate on the centerline of their respective restraining block embedments. (See Figs. 46 and 47).

Note: Placement at this point will be easier then later during alignment of the accessory base. Wires may be attached to the support guide and plate so they can be pulled out if an interference occurs when setting the accessory base.

4. While on blocking, clean, debur and hone the accessory base sole plates.
5. Carefully lower the accessory base over the foundation bolts and onto the sub-sole plate shim packs. (See Fig. 10).

   Refer to MLI 1603 – Foundation Bolting arrangement, MLI 0306 – Outline Mechanical Gas turbine and Load, and MLI 0323 – Outline, Foundation Interface-Gas turbine Package, for Shim pack thickness and calculations if necessary.
6. Remove the plastic pipe from anchor bolts. (See Fig. 5).
7. For initial positioning, align the factory punch marks, scribed on the ends of the accessory base to the line struck from the centerline of the foundation hold-down bolting.
8. Move the accessory package into alignment with the centerline of the foundation.

CAUTION: Move the base as a rigid body to minimize axial flexing. Jacking should be done in close proximity to the base lifting trunnions.

9. Assemble the nuts and washers on the anchor bolts and tighten to the maximum torque. (See Table 2).

10. Check the torque periodically, as with the generator.



Figure 10. Lowering Accessory Base on Foundation

Setting the Gas Turbine Base

1. Prior to setting the turbine base, the mechanical centerline should be scribed across the forward and aft restraining block embedments.

2. Verify that the gas turbine rotor lock (painted yellow), mounted at the No. 2 bearing area, is tightly bolted. (See Fig. 11).

3. Remove and discard the 3/16 inch steel shipping cover which is tack welded to the aft, lower end of the turbine. (See Fig. 12).

CAUTION: When lifting the turbine base, do not allow the base to be out of level more than six (6) inches over the entire length of the package.

4. Place the forward turbine support guide and the aft turbine support plate on the centerline of their respective restraining block embedments. (This will make it easier later during alignment of the turbine base. Wires may be attached to the support guide and plate so they can be pulled out if an interference occurs when setting the turbine). (See Figs. 14 and 15).

5. While on blocking clean, debur and hone the turbine package sole plates (equipment support pads).

6. Position the turbine base over the anchor bolts (See Fig. 13).

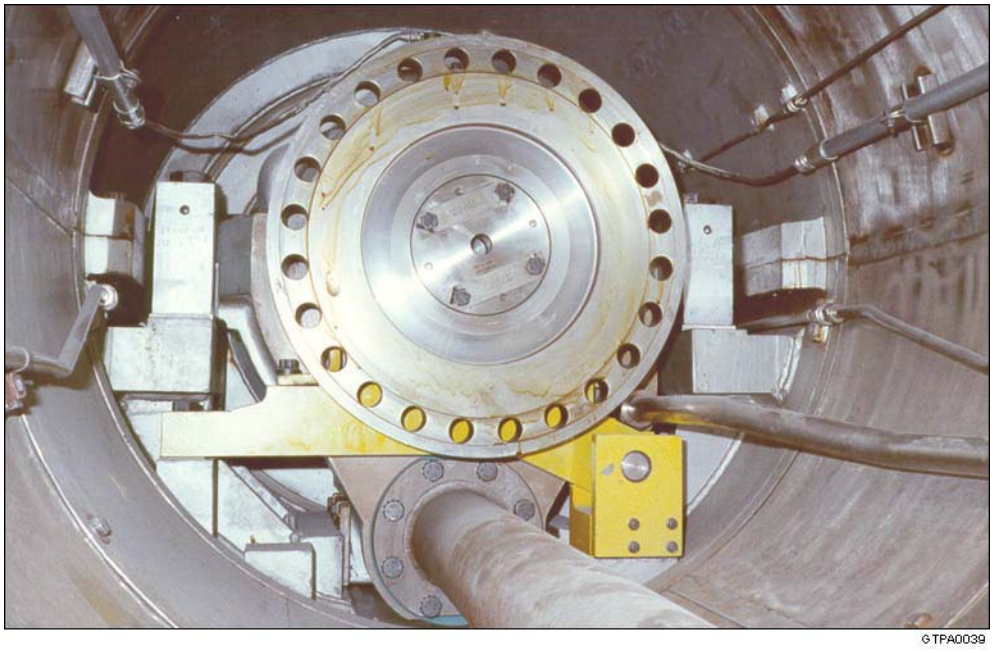


Figure 11. Gas Turbine Rotor Locking Device

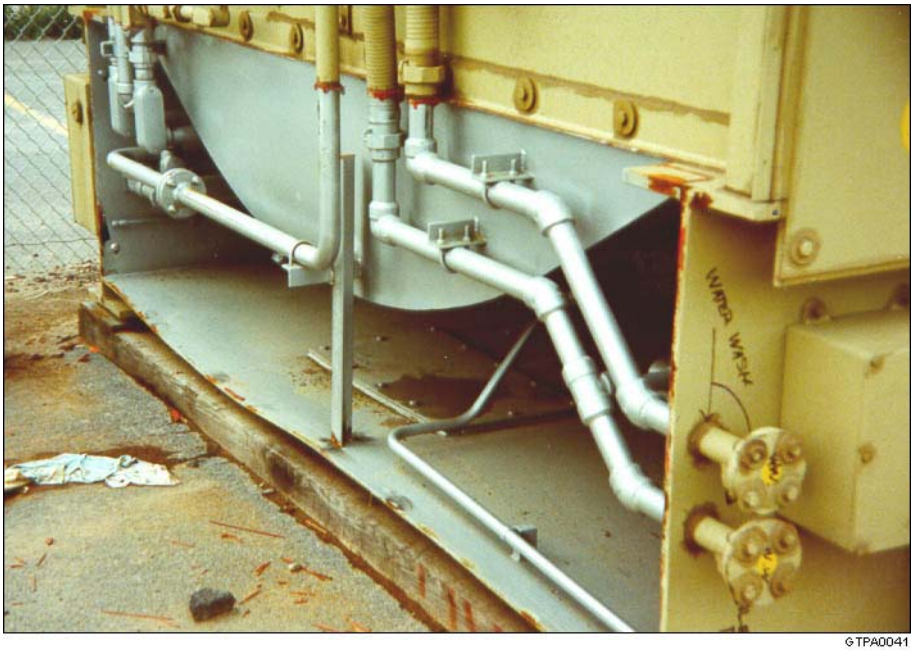


Figure 12. Removed 3/16 Inch Steel Shipping Cover



Figure 13. Lowering Turbine Base on Foundation

7. Lower the turbine package to within several inches of the shim packs.
8. Remove the centerline guide access covers from the base "belly pan".
9. Position a qualified person in the turbine package to verify that no interference occurs between the turbine's centerline foot and the foundation centerline support arrangement during the final setting of the turbine package.
10. If interference is encountered, remove the support guide/plate for machining at a later date.
11. Lower turbine package onto the shim packs.

    Refer to MLI 1603 – Foundation Bolting arrangement, MLI 0306 – Outline Mechanical Gas turbine and Load, and ML0323 – Outline, Foundation Interface-Gas Turbine Package for Shim pack thickness and calculations if necessary.
12. For initial positioning, align the factory punch marks, center punched on the ends of the turbine package, to the line struck from the centerline of the foundation hold-down bolting. The punch marks will be found near the center of the bottom base cross beam. Package punch marks indicate the I-beam centerlines only, not the shaft centerline. (See Figs. 14 and 15 for the jacking trunnion which can be made to assist in moving the base).
13. Remove the protective plastic pipe from the anchor bolts (Fig. 5), and move the package into alignment with the centerline of the foundation. Position the turbine base forward against its anchor bolts so that proper coupling axial spacing can be obtained.
14. Move the base as a rigid body to minimize axial flexing. Jacking should be done in close proximity to the base lifting trunnions.
15. Assemble the nuts and washers on the anchor bolts (Fig. 5), and torque to the maximum torque per the schedule in Table 2. Check torque periodically, as with the generator. Typically, the turbine anchor bolt washers are at least one (1) inch thick.

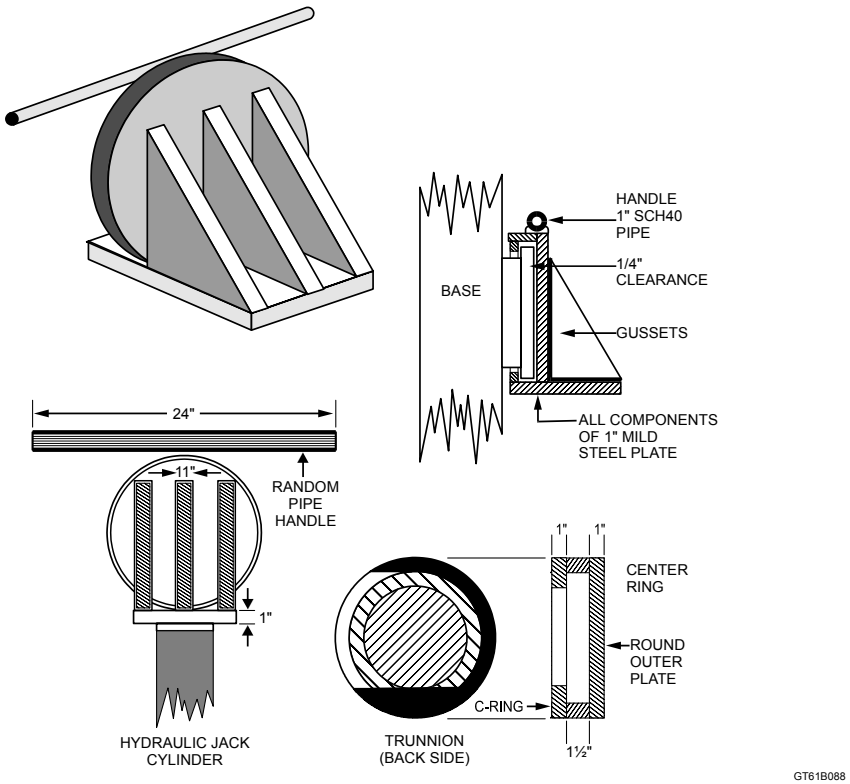


Figure 14. Trunnion Jacking Post Tool

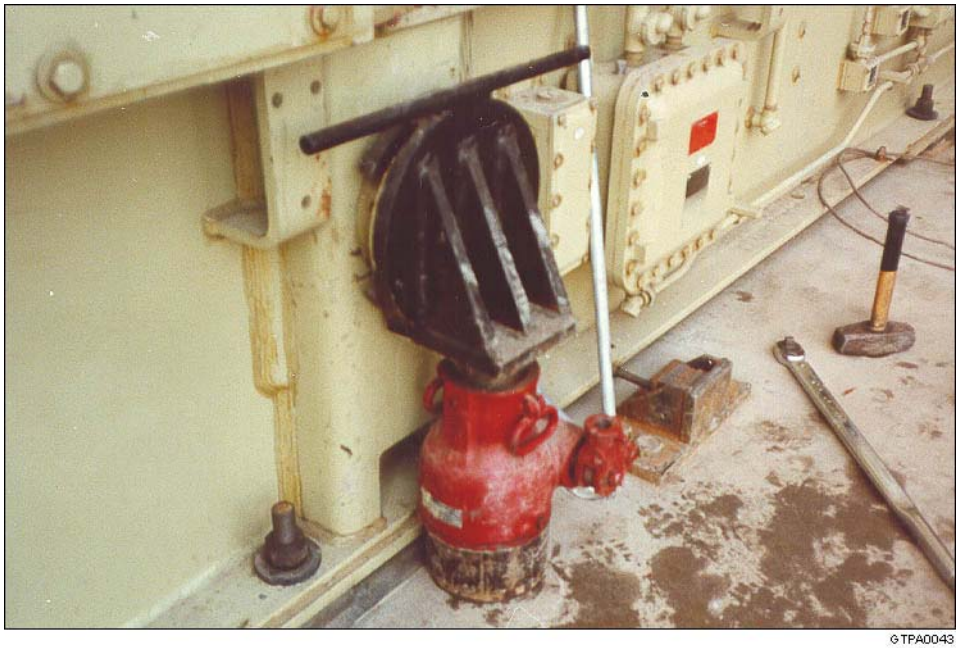


Figure 15. 100 Ton Mechanical Jack and Trunnion Jacking Tool

Preparations for Setting the Load Gear

Extreme care must be taken when setting the load gear to insure that it meets leveling requirements:

- Ensure that the shim packs have been prepared and set on the load gear sub-sole plates. Refer to EM 5036, MLI 1603 – Foundation Bolting arrangement, MLI 0306 – Outline Mechanical Gas turbine and Load, and MLI 0323 – Outline, Foundation Interface-Gas Turbine Package for Shim pack thickness calculations. (See Fig. 16).
- Verify that there are no soft shim packs, i.e., more than six (6) shims in a pack.
- Distribute the shims evenly under all mounting pads to equalize the support load and to avoid distortion of the housing and localized high stresses.
- Remove all protective shipping covers from the load gear.

Note: The leveling requirements of the load gear sub-sole plates and shim packs are held to a closer tolerance than other package supports. All shim packs must be in the same plane within 0.001 inch and level within 0.003 inch/foot in all directions.

If the turbine and generator have been set previously (as described above), the following procedures must be carried out ***prior to*** **setting the load gear** due to the lack of space between packages once the load gear is set.

- Use a solvent to remove the preservative from the shaft extensions, the exterior machine surfaces, the coupling flange faces and the rabbets.
- Inspect the coupling bolt holes, flange faces and rabbets and smooth or remove any nicks, burrs, or other raised surface blemishes with a stone.
- Use a micrometer to measure the load gear hub rabbit fits. Take four measurements every 45 degrees and average to find each dimension. (See Fig. 9 – Rabbet Interference Fit Measurement).
- See the manufacturer's recommendations for lifting and setting the load gear. Rigging should ensure that the gear is maintained in a level position at all times.

Setting the Load Gear

1. While on blocking, clean, debur and hone the load gear base plate (equipment support).
2. Position the load gear over the anchor bolts and carefully lower the load gear into place over the anchor bolts. (See Figs. 17 and 18).
3. Remove the plastic pipe from the anchor bolts.
4. Loosely install the nuts and washers on the anchor bolts.
5. Move the load gear onto the mechanical centerlines.
6. Use a feeler gauge to determine if the shims have a minimum of 80 percent contact.
7. Check centerline position and torque anchor bolts. Clean horizontal joint on both sides in preparation for checking level.



Figure 16. Load Gear Shim Packs Stetting on Sub-Sole Plates



Figure 17.
Lowering Load Gear on Foundation

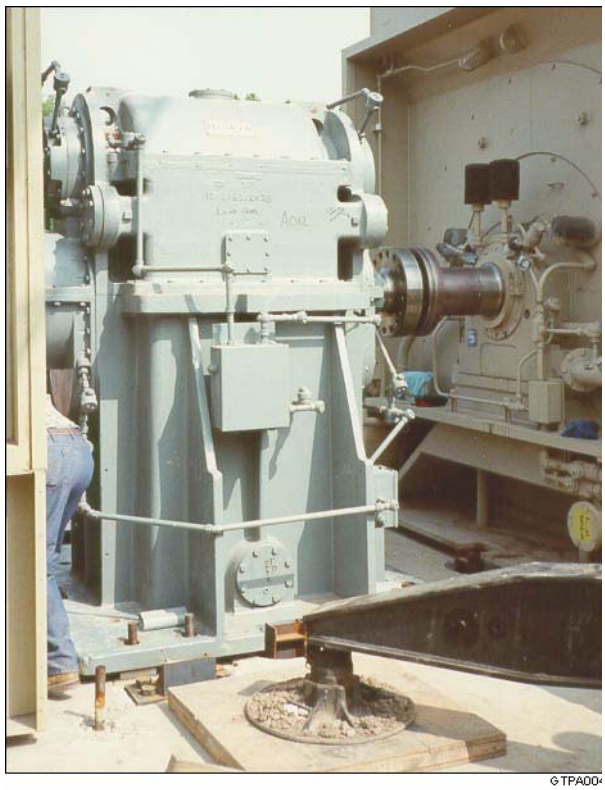


Figure 18.
Load Gear Set On Shim Packs

8. Use a machinist's level to take a level reading axially and laterally. Compare readings on both sides of the Load Gear for repeatability.

9. Make necessary shim changes to level Load Gear to 0.000 inches per foot in all directions. Check for soft foot and 80 percent contact.

10. Again check centerline position, torque anchor bolts and recheck level. Add or subtract shims if necessary.

11. Continue process until Load Gear is level, has no soft foot, sits on the machine centerline with 80 percent contact with all anchor bolts torqued. Reference MLI 1603 for Load Gear anchor bolt Torque value.

ALIGNMENT OF MAJOR EQUIPMENT

Alignment should be accomplished with reference to MLI 0410 (Factory Alignment Instructions) and MLI 0411 (Basic Field Alignment Instructions).

Alignment procedures for the load gear, generator, turbine and accessory base are detailed in the following paragraphs. The overall alignment of the centerline packages (i.e., turbine, generator, load gear and accessory gear) is accomplished without rotating either the turbine or generator shafts. Alignment readings are obtained by mounting indicators on the load gear hubs or accessory gear hub and sweeping the stationary turbine or generator coupling hub. In preparation for these readings, the hub surfaces must be cleaned and inspected to insure the coupling bolt holes, flange faces and rabbets are free from nicks, burrs or other raised surface blemishes. (Refer to EM 5211 – MS 6001B Load Gear Field Inspection; Load Gear Disassembly and Alignment Readings section).

Rough alignment is achieved using a straight edge to roughly align the generator to load gear coupling. The load gear should remain in its fixed position, and the turbine and generator are aligned to it. Prior to the start of rough alignment the bases should remain torqued to the anchor bolts at maximum value to minimize any distortion. (See Table 2 for torque values).

Note: Feeler gages and an alignment fixture, needed for the rough alignment between the turbine and load gear.

Initial alignment is performed with dial indicators and an alignment fixture (in a manner similar to final alignment), but prior to installation of inlet and exhaust systems to allow easier movement of turbine.

Final alignment of the turbine must be accomplished after the weight of the inlet duct elbow and the exhaust duct/stack are placed on the turbine.

Load Gear Alignment Preparation Activities

1. Check the level of the load gear to ensure that it has remained at 0.000 inches, in both axial and transverse directions.
2. Check the torque on load gear anchor bolts to ensure that it has remained at 1,000 foot-pounds.
3. Ensure that the load gear is set on shim packs with a minimum of 80 percent contact at each hold-down point.
4. Remove the load gear shipping lock.
5. Ensure that the low-speed quill shaft centering ring (MLI 0411) is in place for final alignment. (See Figs. 19 and 20).

Note: This ring must be removed after final alignment to prevent damage to the equipment.

6. Prepare a special wrench to fit load gear hub to facilitate shaft rotation. (See Fig. 21).
7. Prepare and assemble thrust collars for use on both the turbine and generator end hubs of load gear to minimize float during rotation. (See Fig. 22).
8. Assemble lock keys on both sides of the H-beams on all four sides of the load gear base. (See Fig. 23).
9. Ensure that the lock key total clearance to the H-beam is in accordance with the foundation drawings/alignment specification. (The typical clearance is 0.000 to 0.002 inches, but may be unit specific).
10. After the lock keys are set, tack weld the bolt head to the washer, then the washer to the key.
11. Drill two dowel holes in each key and in the load gear base for insertion of dowel pins after final alignment.

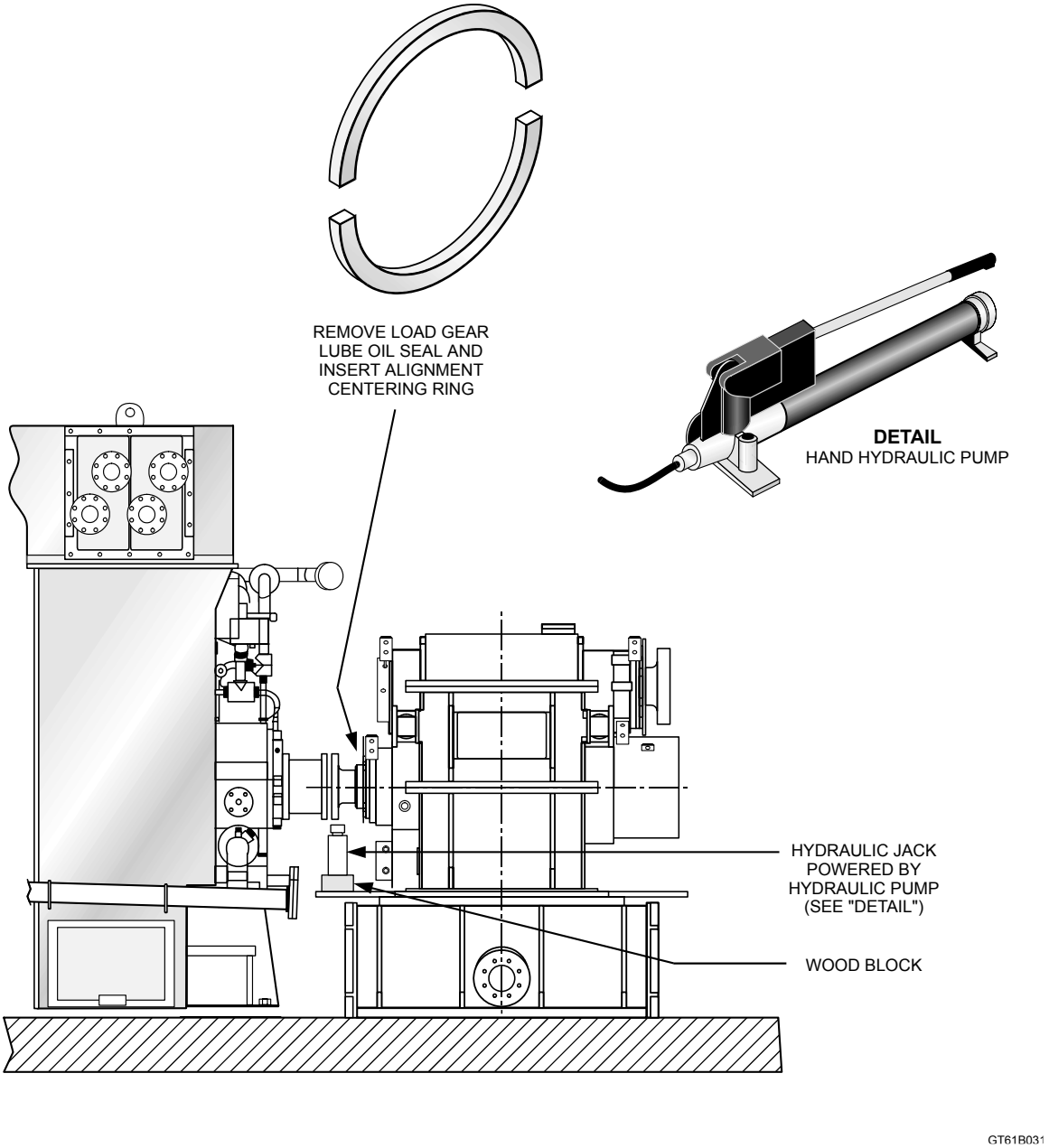


Figure 19. Centering Ring Installation Arrangement

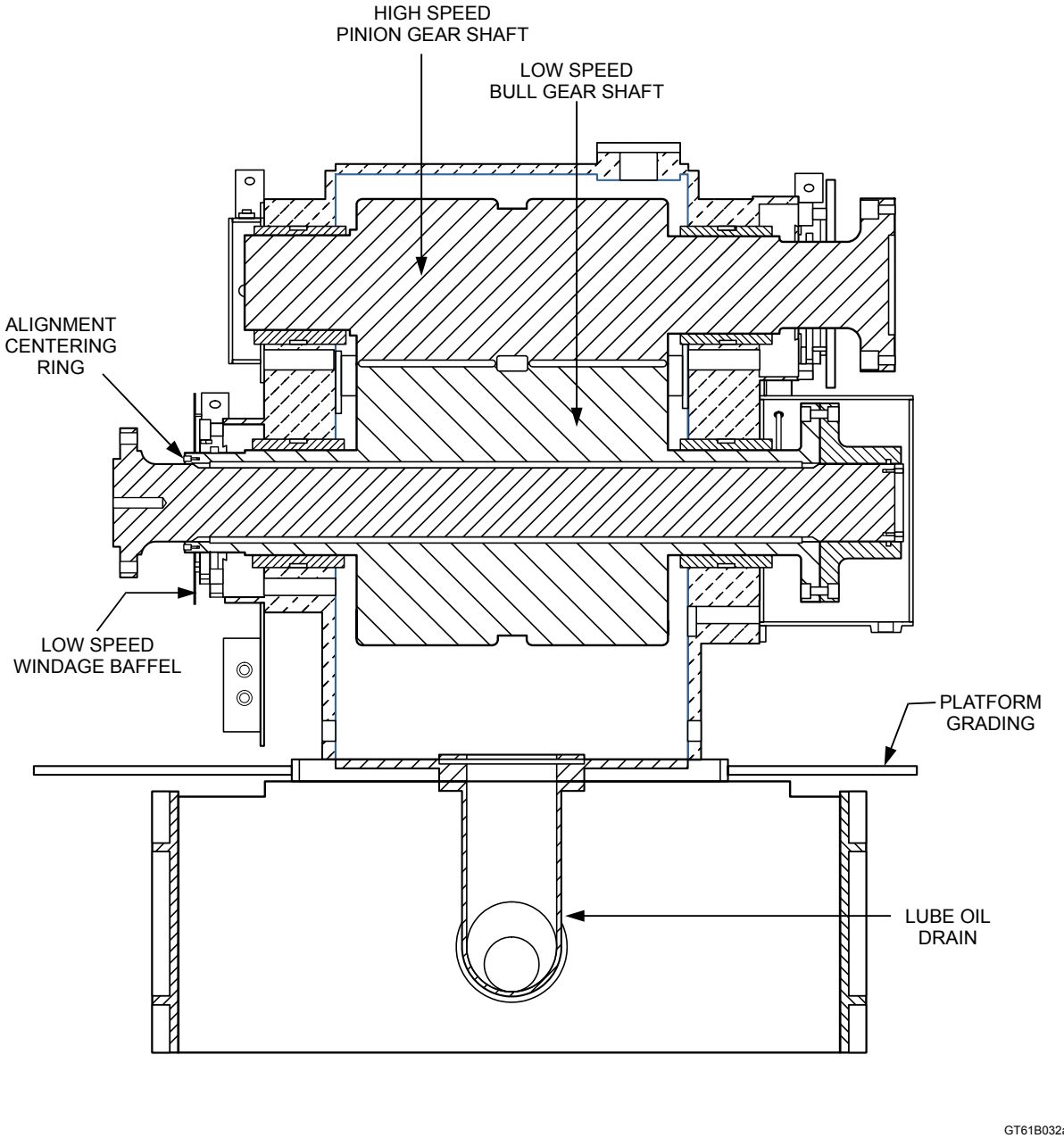


Figure 20. Centering Ring Installation Arrangement Detail

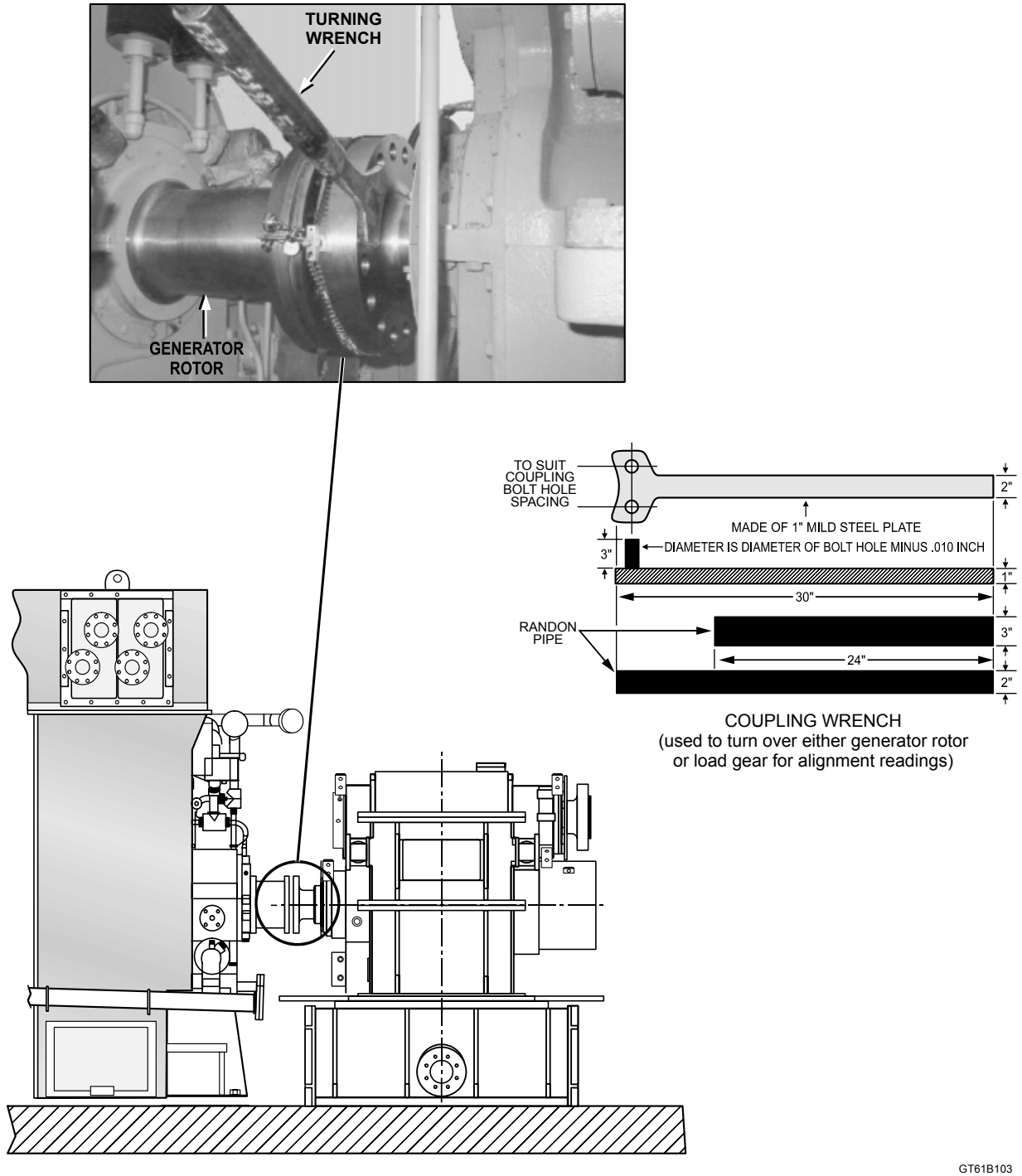


Figure 21. Fabricated Coupling Wrench for Turning Load Gear

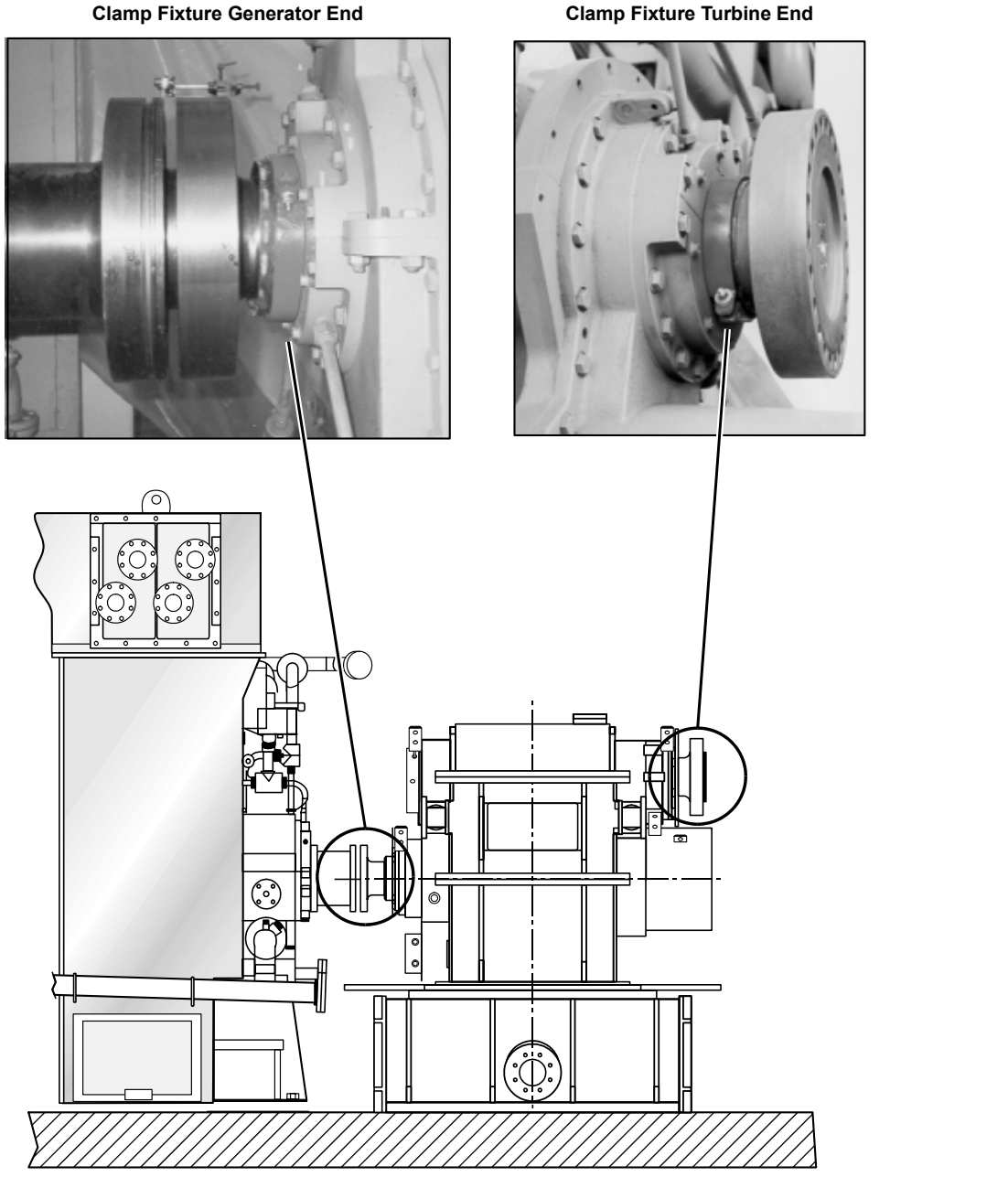


Figure 22. Bull and Pinion Gear Shaft Fixture Installation

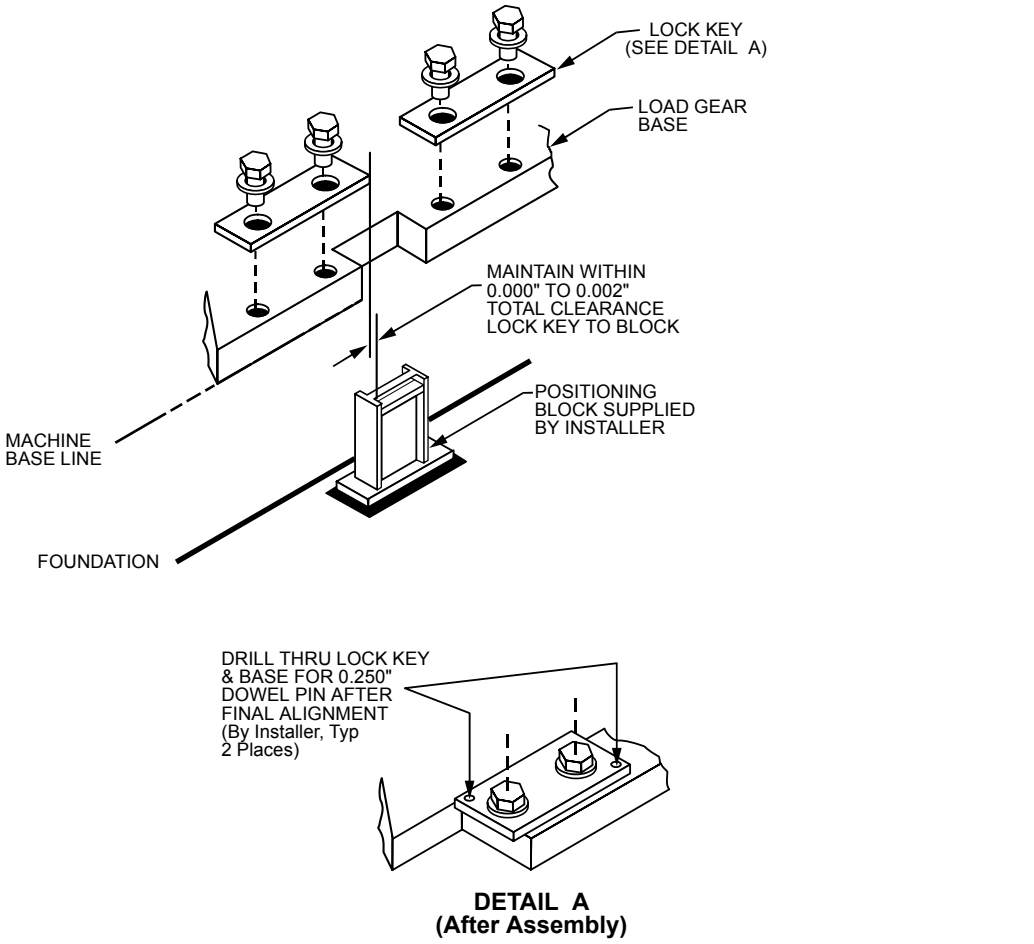


Figure 23. Lock Key Installation Details

Temporary Load Gear Lubrication

A method for providing lubrication to the load gear during alignment rotation is provided below.

1. Obtain approximately 200 gallons of No. 32 turbine lube oil, a hand pump and clean one inch rubber hose (See Fig. 24, and MLI 0465 – Lube Oil Selection Instructions).
2. Remove the one-half inch pipe plug from the feed pipe flange at the right rear side of the gear box.



Figure 24. Setup for Lubricating Load Gear

3. Install a reducer and a one inch pipe nipple.
4. Attach a rubber hose at the nipple in order to pump lube oil into the load gear prior to turning for alignment.
5. Remove the inspection plate from the left side of load gear and inspect the teeth of the pinion and bull gear, making sure that they are free and clear to rotate.
6. Pump some turbine lube oil into the load gear until it comes through the shaft bearings inside the gear box. Do this each time prior to rotating the gear.
7. Rotate the gear and inspect the teeth for cleanliness and signs of distress. Check for any signs of corrosion, damage or foreign objects inside.
8. Replace the inspection plate on the load gear. If the load gear is left sitting for a long period of time before operation, lubricate periodically to prevent corrosion.

Alignment of Generator to Load Gear

1. Ensure that the generator coupling flange is at the proper axial location with respect to the centerline of the generator bearing, minus 0.750 inch (as shown in Fig. 8).
2. Inspect the coupling bolt holes, flange faces and rabbet fits, then clean, smooth or remove any nicks, burrs, or other raised surface blemishes with a stone.
3. Mark the generator coupling and load gear shaft hubs into quarters with a marking pen.
4. Mount a dial indicator on the load gear hub and read to the generator hub. (See Figs. 25 and 26).
5. Ensure that the coupling flange rims are concentric within the alignment specification. (Typically 0.002 inches).
6. Turn the load gear hub using the special wrench mounted on the turbine end flange of the load gear. (See Fig. 21).
7. A dial indicator should be mounted to the back face of the gear hub to record actual float. This reading must be added to or subtracted from the face readings. (See Fig. 25).
8. Take a rim reading and a four-point face reading at each 90 degree turn.
9. Ensure that the indicator is returned to zero after each turn. One complete revolution will yield a sixteen-point face runout and a four-point rim runout. (See Fig. 27 for typical alignment specifications).

NOTE: During alignment, make sure that the generator rotor is maintained at its cold alignment dimension, minus 0.750 inch, by maintaining a constant measurement from the center punch marks on top of the generator bearing to the end of the hub.

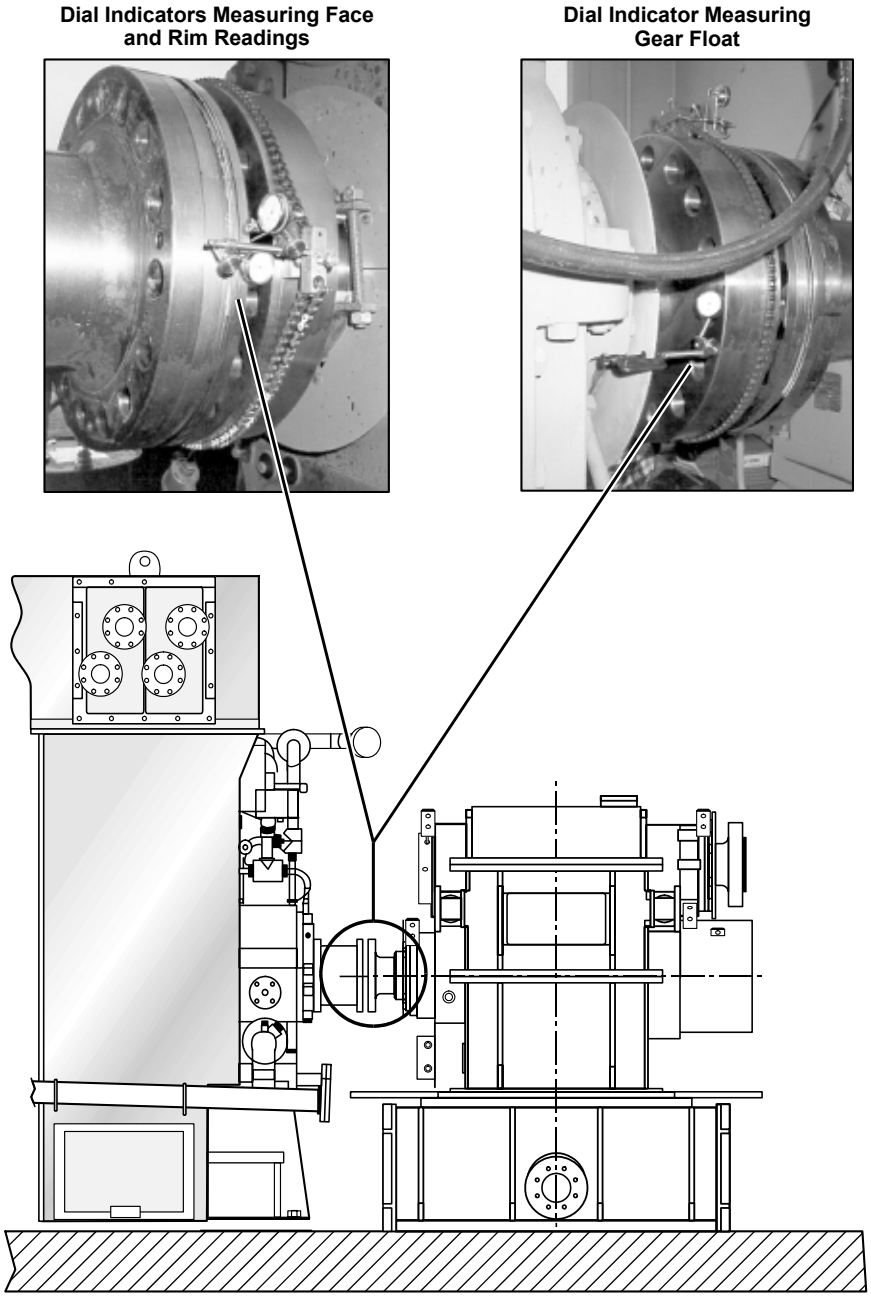


Figure 25. Indicator Arrangement for Rim Readings

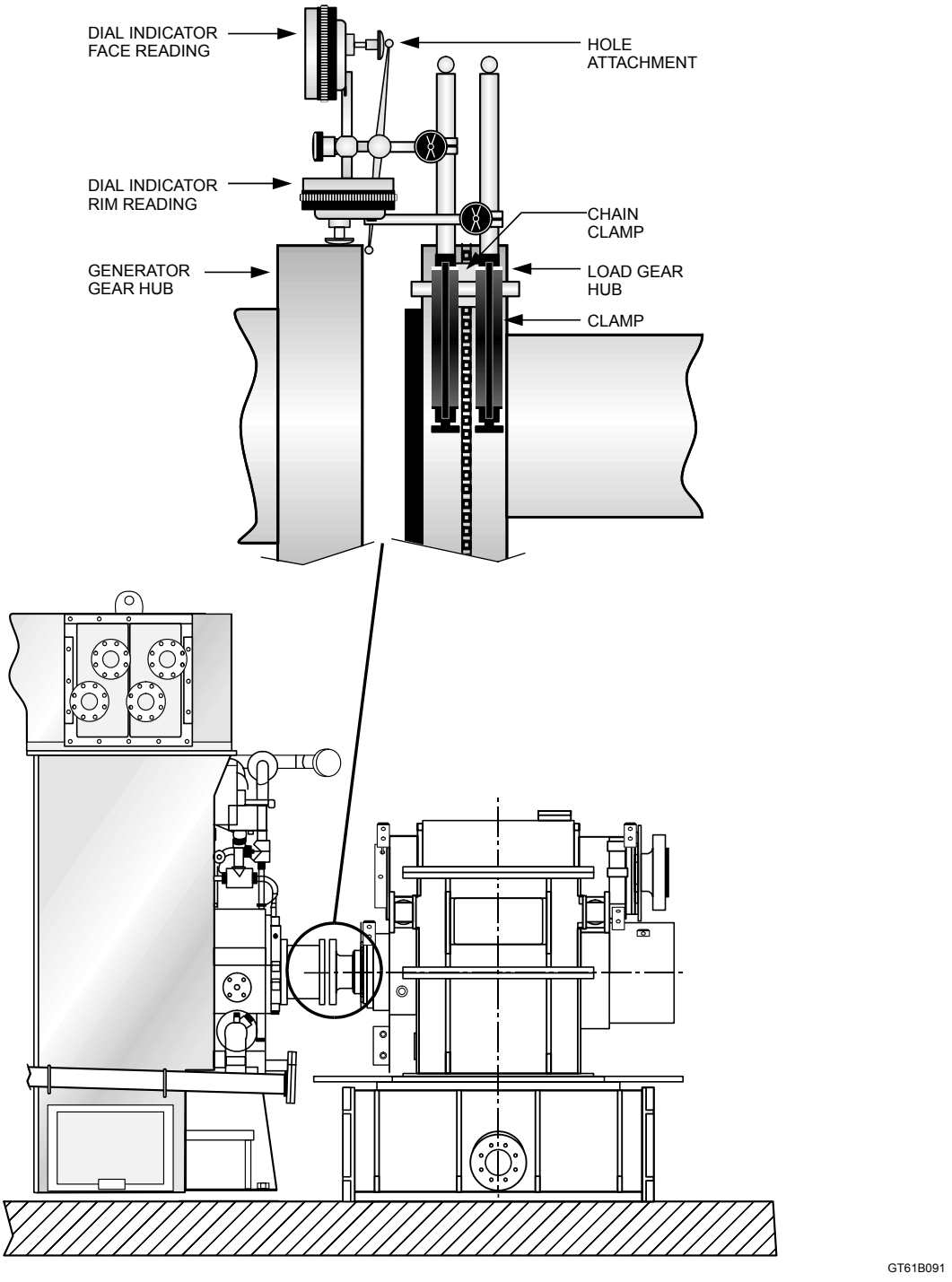


Figure 26. Indicator Setup for Rim Readings

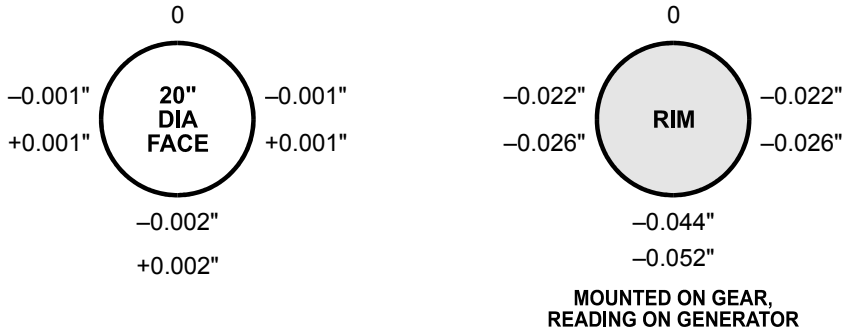


Figure 27. Typical Load Gear Alignment Readings

Notes: The rim readings will be reversed in sign if dial indicator on generator reading gear. The values give are typical. Refer to alignment specifications for site specific values.

10. During this alignment procedure, all moves made to bring the load gear/generator into alignment must be made to the generator. **Do not move the load gear.**

11. When making side-to-side moves to achieve transverse alignment, care should be taken not to "overshoot" the move that is required.

Note: The package generally does not move smoothly but "jumps", and it is difficult to control the actual movement. The "jumping" movement can be minimized by placing shims between the jacking post and base on the side of the base opposite to the jacking post being used to move the base, leaving a clearance equal to the amount of movement required.

12. Recheck the alignment readings for both concentricity and parallelism of the coupling flanges after every shim adjustment.

    Do not make up the generator coupling at this time. The load gear has to be free to turn in order to take the turbine-to-load gear alignment readings.

Alignment of Turbine to Load Gear

1. Inspect the coupling bolt holes, flange faces and rabbets and clean, smooth or remove any nicks, burrs, or other raised surface blemishes with a stone.

2. Remove the manway cover plate on the right side of the aft end of the turbine base.

3. Enter the manway and mount a 25-ton jack on the jacking fixture provided inside the turbine base.

4. Jack against the aft turbine shell vertical joint.

5. Round off one end of a three inch piece of "All Thread" (used by Generator to lock the field during shipping). Cut it to fit between the top of the jack and the turbine shell flange to allow a sideways movement (left to right) of the turbine shell.

6. Jack up the turbine just enough to back out the shipping pin bolts (one on each side of the gib block). This will be about 0.030 inches. (See Fig. 28).



Figure 28. Removal of Shipping Pin (with turbine jacked up)

7. Conduct a "drop check" on the aft turbine support legs. (Refer to MLI 0411). Typically the check is made as follows:

   a. With the unit on three-point support, as described above, use feeler gages to measure the gaps between the bottoms of each of the two aft turbine support legs and the tops of their bases.

   b. Remove the shim packs that are taped to both support legs and measure and record the thickness of each.

   Note: Each shim pack should consist of one thick and a multiple shim pack.

c. Install these shim packs under their respective aft turbine support legs and, using the measurements taken during Step b, check that the gap variation between the right and the left aft turbine support legs does not exceed 0.015 inch. If the gap variation exceeds 0.015 inch, split the difference between the legs by adding one-half of the required thickness shims to one leg and removing one-half of the required thickness shims from the other leg. Advise Product Service if any difference is observed in this drop check. (See Fig. 29).

Note: Additional shims are provided with MLI 0106 – Installation and Start-up Extras.

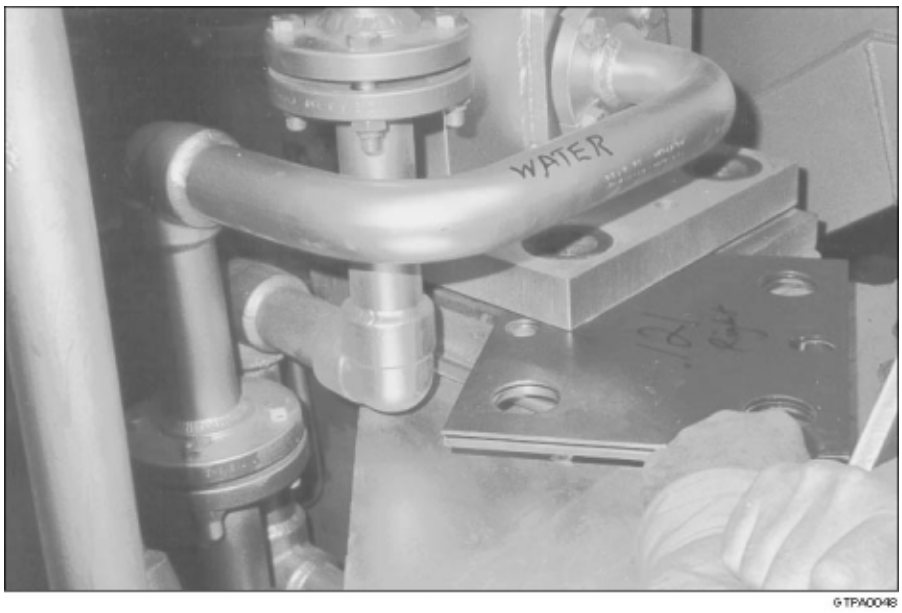


Figure 29. Installing Shims to Turbine Aft Support Leg

d. Before setting the turbine support legs down on the shim packs, drive dowels down through the shim packs into the base pads.

e. Lower the jack so that the turbine is supported by the aft turbine support legs.

f. Torque the four bolts in each aft support leg base pad per specifications. (See Table 2).

g. Assemble new three (3) inch bolts (included in the summary sheet container) into the turbine gib block. Screw both bolts up snug against gib block to prevent movement of Turbine during final alignment. After final alignment is finished, set the clearance in accordance with the site specific alignment specification and foundation interface drawings. (Typically the clearance will be 0.002 to 0.004 inches on both sides). (See Fig. 30).

h. Secure the jacking bolt by tack welding an angle lock from each bolt head to gib block.

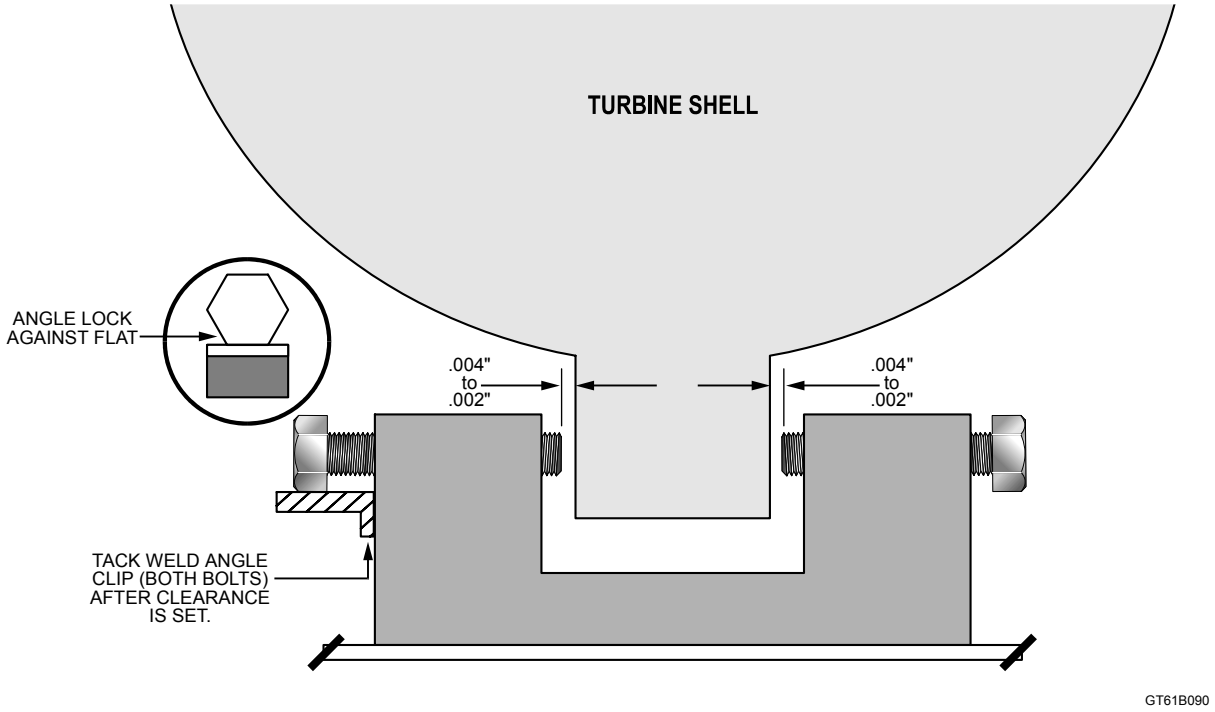


Figure 30. Turbine Centerline Guide Clearances

8. Remove the turbine rotor lock (painted yellow) from the No. 2 bearing area. (See Figs. 31, 32, 33 and 34).

NOTE: Before starting turbine alignment make sure the turbine rotor is thrust forward. This is done by setting a dial indicator at the No. 1 bearing position and thrusting the turbine rotor aft. This places the rotor against the inactive thrust bearing. Jack the rotor forward and measure the bump. Push the rotor aft, and then forward again (measuring the bump each time). Leave it in the forward position against the active thrust bearing. The average of the three bump readings makes the bump check reading. Jacking is done by means of a 25-ton hydraulic jack ("Port-a-Power"), and a wooden block against the load gear and accessory gear hub faces.

CAUTION: **Both the load gear hub and accessory gear hub must be securely blocked between the back of each hub and its casing to prevent internal damage during thrusting. (See Fig. 35).**

9. Mark the turbine rotor hub into quadrants (quarters) with a marking pen. (Refer to MLI 0411 – Basic Field Alignment Instructions for details).

10. Assemble the turbine alignment fixture to the load gear hub. (See Fig. 36).

NOTE: This fixture is mounted on the load gear hub so that the dial indicator readings can be taken on the turbine rotor hub.

11. Assemble a special coupling wrench to the generator load gear hub and turn the gear over to take a sweep with the dial indicator on the turbine rotor hub. (See Fig. 21).

12. Face and rim readings can be taken at the same time with two dial indicators. A dial indicator must also be mounted to the back face of the gear hub to record actual float. This reading must be added to or subtracted from the face readings. At all times ensure that the distance from the gear hub face to the turbine hub face is 76 inches plus or minus 0.000 inches. (Refer to MLI 0411).

13. Adjust the shim packs under the turbine base to obtain the turbine-to-load-gear vertical alignment.

14. Adjust the horizontal alignment using jack posts on each side of the turbine in the foundation. (See Fig. 15)

15. For a side-mounted exhaust, assemble the exhaust plenum side cover (wing) and weld it in place (See Fig. 37). Ensure that all six (6) gussets are welded on the front and aft flanges (Refer to EM 5055 – Exhaust Duct and Stack).

    Assemble the top cover to the exhaust plenum and assemble the expansion joint. Do not bolt the expansion joint to the HRSG or exhaust duct at this time. (See Fig. 38).

16. For a straight up flow exhaust system, assemble both exhaust plenum side covers, the transition duct, silencer and stack segments. During stack segment assembly check, plumb each segment as it is installed. Check welds and insulation at segmented joints. Check distances between segments where expansion joints are to be assembled are within specification. Conduct light check for leaks and clean exhaust plemum before operation. (See Fig. 39).

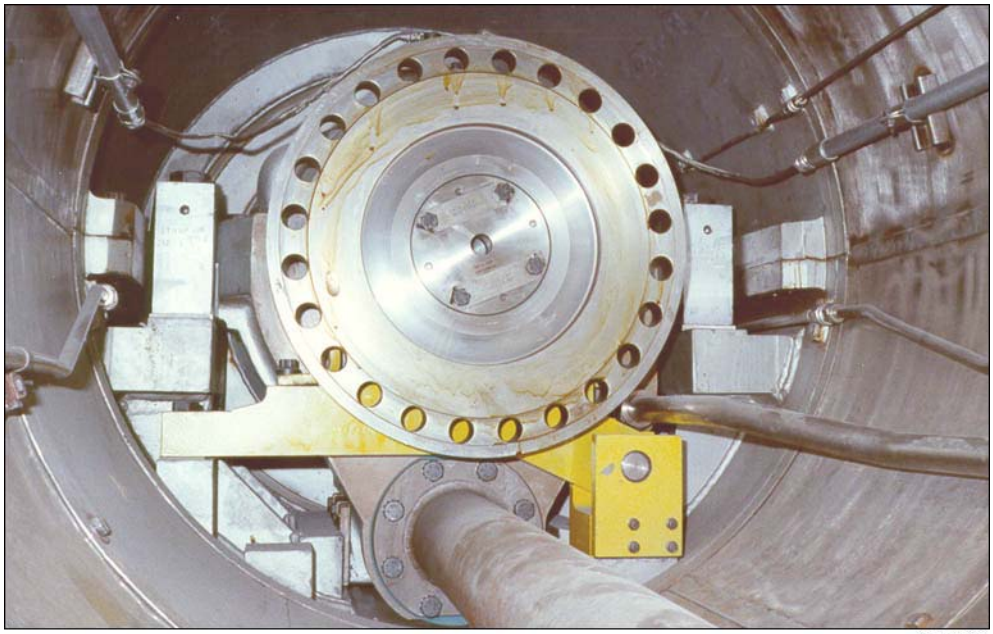


Figure 31. Rotor Locking Device Assembled by Factory for Shipping



Figure 32. Rotor Lock Being Disassembled

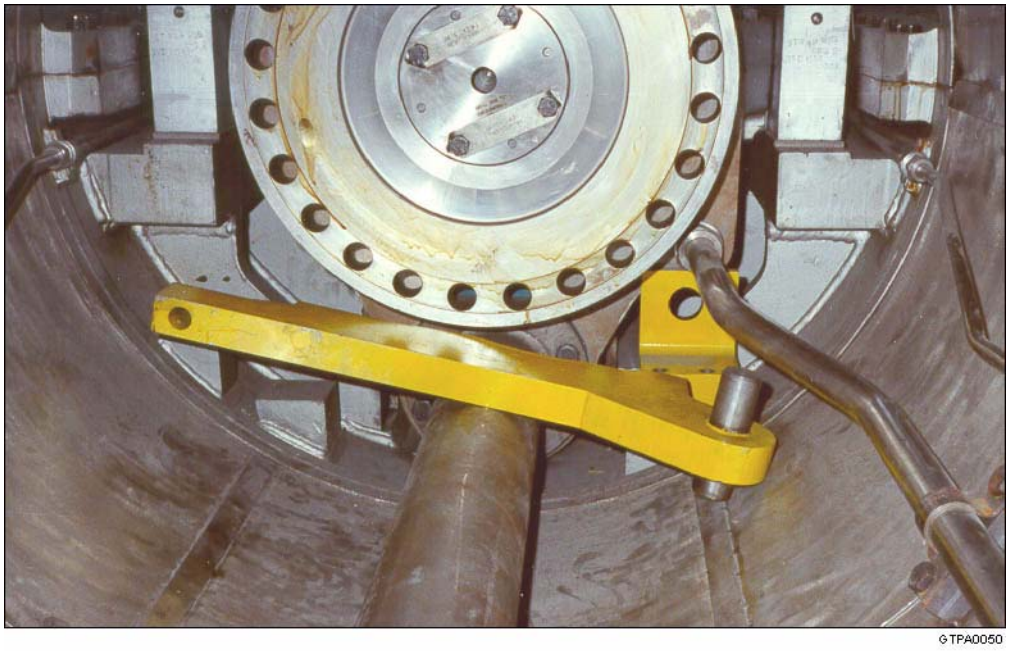


Figure 33. Rotor Lock Partially Disassembled

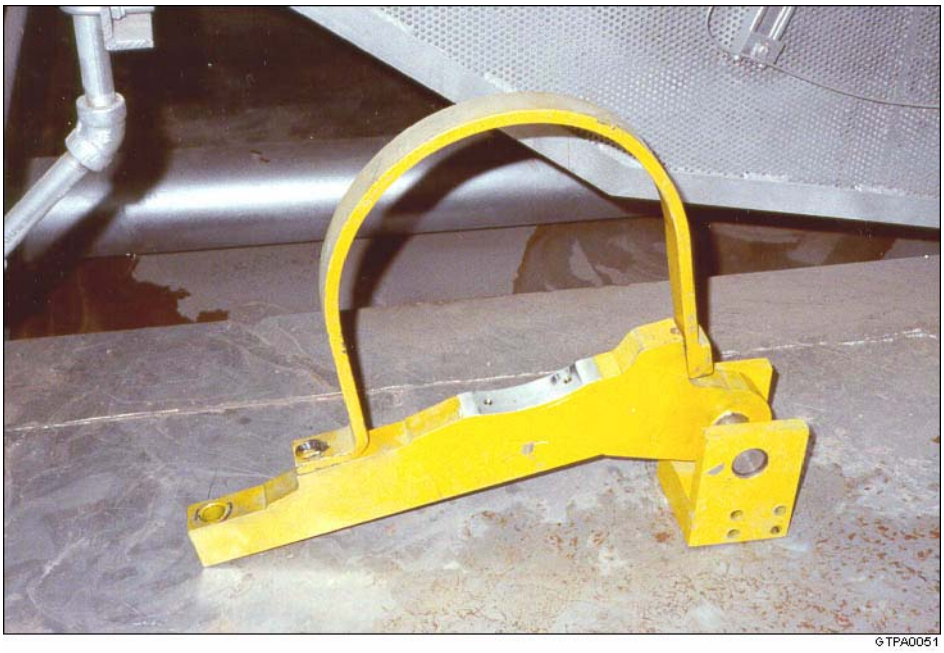


Figure 34. Rotor Lock Removed from Turbine Rotor

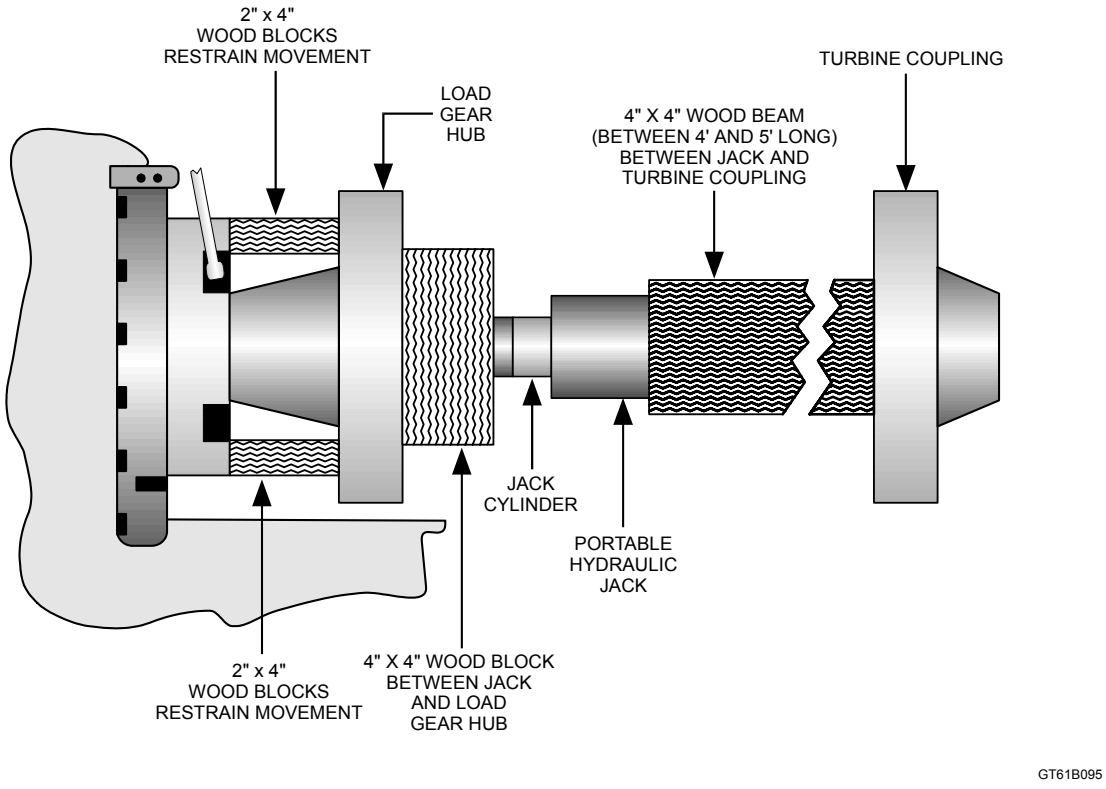


Figure 35. Load Gear Hub and Accessory Gear Hub Jacking Arrangement

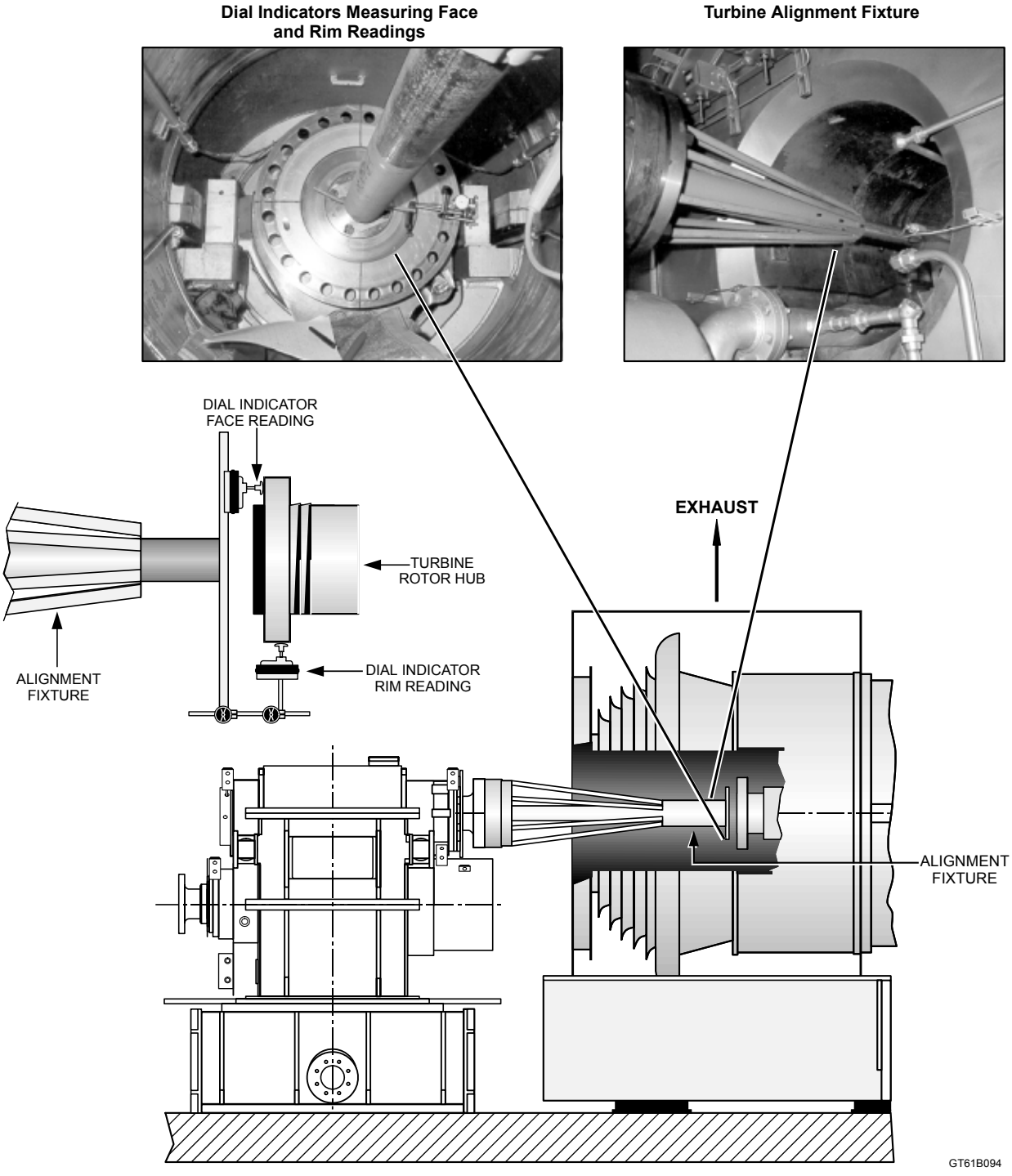


Figure 36. Turbine Alignment Fixture with Dial Indicator Assembly



Figure 37. Rigging Exhaust Plenum Side Cover (Wing)



Figure 38. Rigging Top Cover of Exhaust Plenum

17. Assemble the turbine inlet side wings, transition piece and elbow. (See Figs. 40, and 41). Refer to EM 5051 – Inlet Ducting Installation and Maintenance. Verify gasket is installed between all duct sections where necessary. Torque all duct sections to specified torque value. Clean all duct sections and conduct light check for leaks on clean side. Install Filter elements after cleaning is complete.

    NOTE: The inlet and exhaust duct pieces are assembled at this time to impose the full weight of these pieces on the turbine while watching dial indicators to see if the weight will change the alignment. If the readings change, adjust the shim packs under the turbine base to obtain the turbine-to-load-gear final alignment.

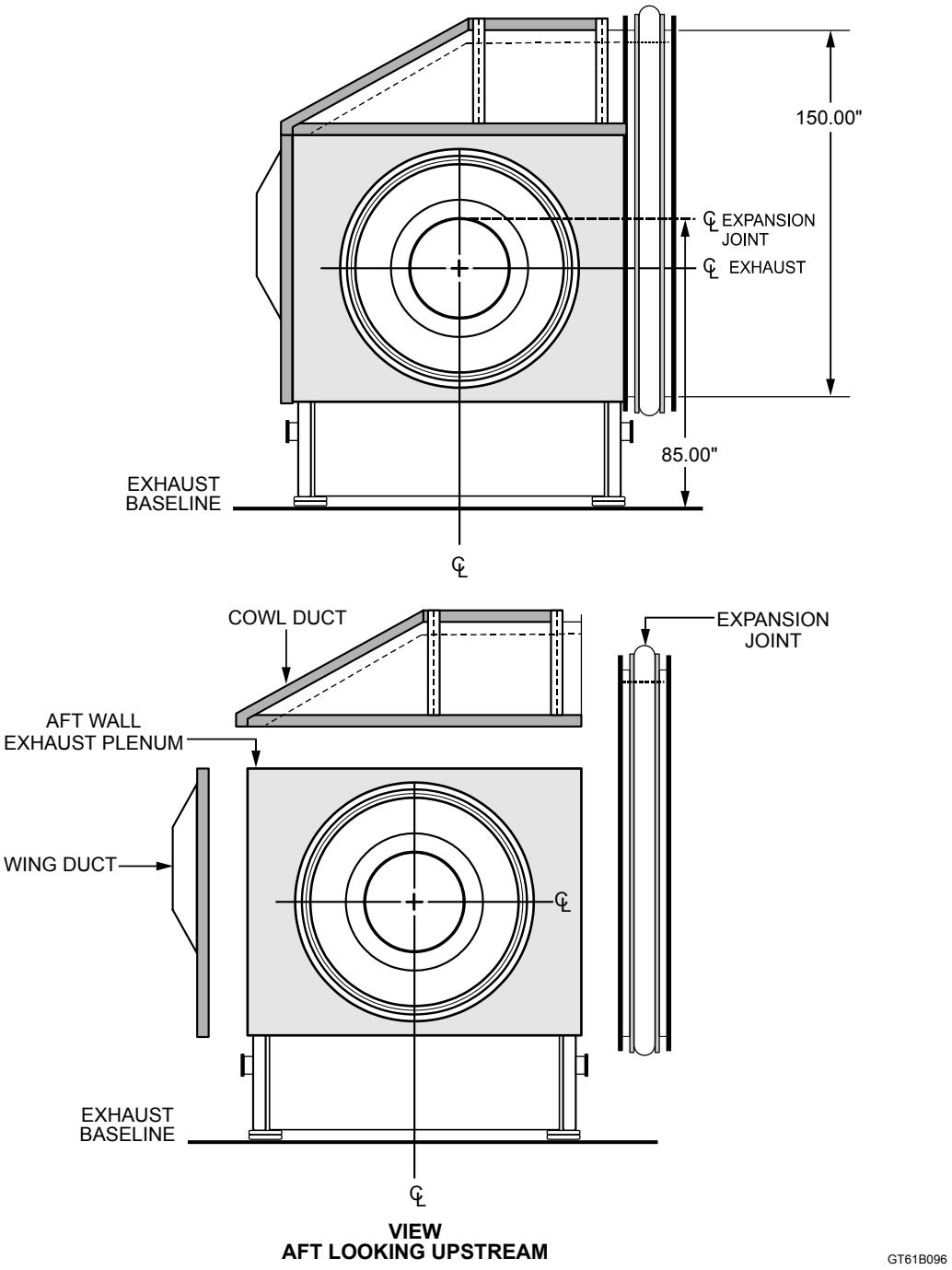


Figure 39. Exhaust Duct Arrangement – Typical Layout (MLI A042)



Figure 40. Assembling Turbine Inlet Wing



Figure 41. Turbine Inlet Wing

18. During the assembly of the inlet and exhaust ducting, the third stage buckets and inlet guide vanes should be inspected, and a good visual inspection should be made to insure that all shipping papers and foreign material are removed from these areas.

    NOTE: This step also provides an opportune time to rotate the turbine rotor and listen for rubs.

    a. Assemble the turbine rotor turning fixture (fabricated by the installer, see Fig. 42), to the front of the turbine rotor using four of the special bolts provided for the accessory coupling.

    b. Place one millwright in the inlet plenum, one in the exhaust plenum, and two more along side the horizontal joint of the compressor case and the turbine shell. (The two stationed along side the unit should use sounding rods to help listen for internal noise.)

c. Rotate the turbine rotor 180 degrees while listening for internal noise.

d. Since the turbine rotor bearings are not being lubricated at this time, it is important to rotate the rotor the absolute minimum amount.

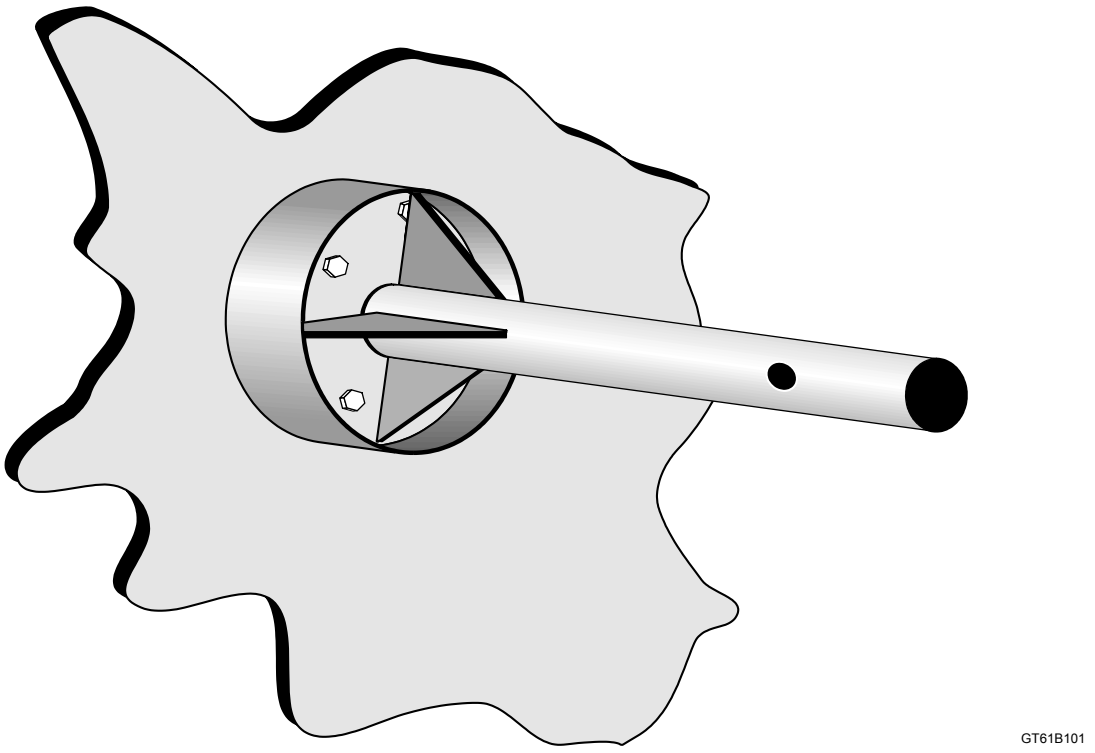


Figure 42. Turbine Rotor Turning Fixture

Alignment of Accessory Base to Turbine

Refer to the Basic Field Alignment Instruction MLI 0411, and proceed as follows:

1. Clean the turbine rotor hub at the No. 1 bearing and the accessory gear hub.
2. Measure and record the rabbets in both the turbine and the accessory gear hubs.
3. Remove the cover over the jaw clutch from the starting means torque converter to the accessory gear. (This will allow the accessory gear to be turned over to take readings. The accessory gear is very easily turned over for alignment runouts.)
4. Assemble the alignment fixture and bolt it to the accessory gear. (See Fig. 43).

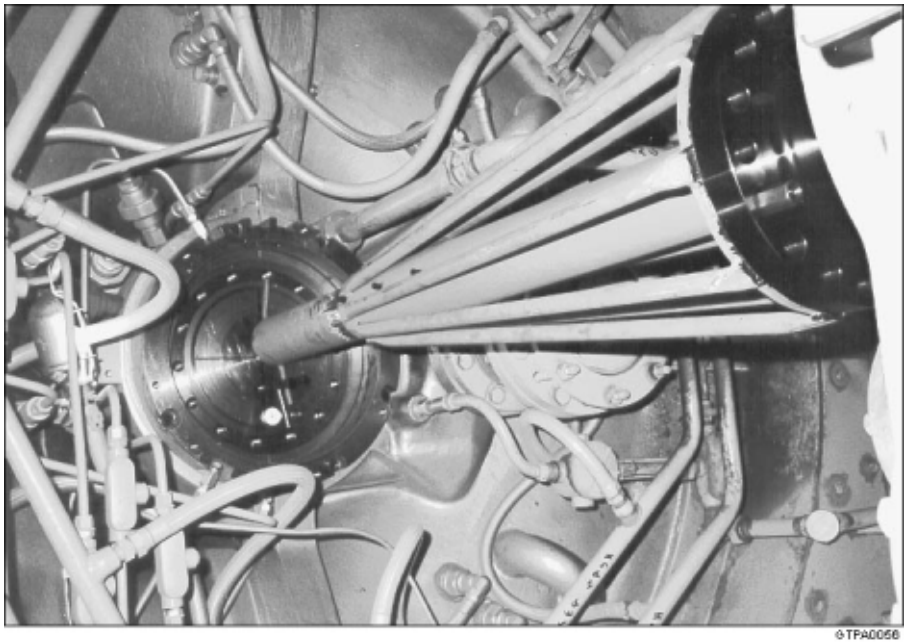


Figure 43. Alignment Fixture Mounted on Accessory Gear

5. Assemble two dial indicators to the fixture and take rim and face readings at the same time on the turbine rotor hub.

NOTE: When taking a face reading at each quarter mark, place a bar behind the gear hub and move it toward the turbine to thrust the accessory gear forward. If this is not done at each quarter turn, false face readings will result. At all times ensure that the distance from the gear hub face to the turbine forward stub shaft is 52.00 inches plus or minus 0.060 inches. (Refer to MLI 0411).

6. Loosen the anchor bolt nuts and adjust the shim packs under the accessory base, using a special trunnion jacking tool to obtain the accessory gear to turbine vertical alignment per MLI 0411. (See Figs. 14 and 15).

7. Adjust the horizontal alignment using jack posts on each side of the accessory base. (Refer to MLI 0411).

NOTE: Before taking final rim and face alignment readings, assemble the lube oil cooling water heads to the cooler bundles located in the right side of the accessory base. See EM 5080, Field Assembly of Cooling Water System Arrangement.

Installation of Centerline Keys

The load gear centerline keys were installed when the load gear was set, and the remaining packages were subsequently aligned to the load gear. Installation procedures for the Generator, Turbine and Accessory base centerline keys are presented below.

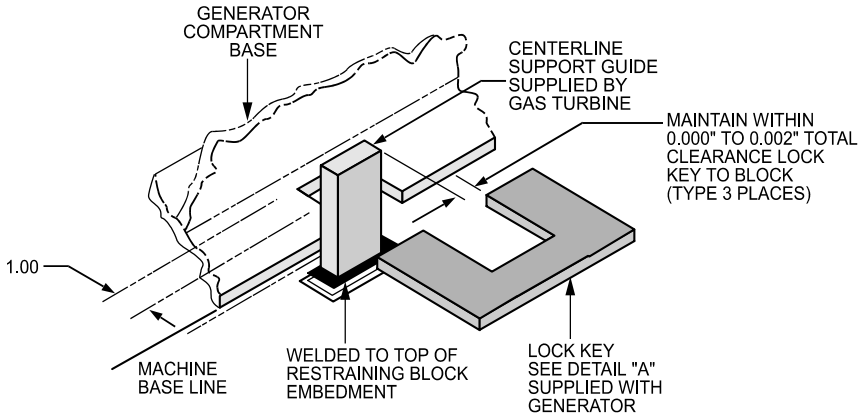
Installation of Generator Keys

1. Assemble the U-shaped keys around the front and aft embedment mounted guide blocks. Check the clearance on both sides to be (typically) 0.000 to 0.002 inches, with an axial clearance of 0.002 inch and weld the keys to the generator base. Refer to MLI 0411. (See Fig. 44).

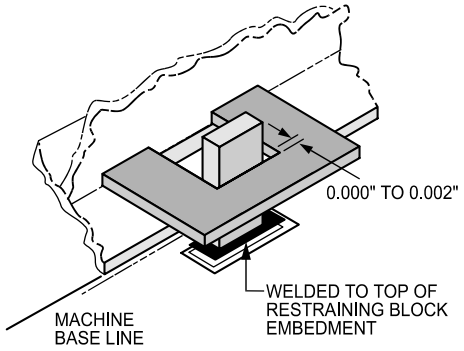
NOTE 1: The location of the bus duct enclosure at the aft end of the generator may not permit welding one side of the U-shaped key. Consideration should be given to locating the key on the bottom side of the base.

NOTE 2: On units currently being shipped, there has been a design change on the generator guide blocks. The new type guide blocks are welded to the upper surface of the restraining block embedment in the same manner that the turbine and accessory base support guide blocks are assembled.

2. Loosen the generator anchor bolt nuts to be 0.010 inch loose, see Generator Mechanical Outline.
3. Lock the anchor bolt nuts in place. If the anchor bolts have to be cut down, leave approximately 1-1/2 inches of thread above the top of the nuts. (See Fig. 45).



DETAIL "A" (BEFORE ASSEMBLY)



DETAIL "A" (AFTER ASSEMBLY)

GT61B093

Figure 44. Generator Lock Keys

Installation of Turbine and Accessory Base Keys

1. Insert the four (4) inch diameter pin (see Fig. 46) through the base mounted centerline support block and into the centerline support plate. (This plate was placed on its restraining block prior to setting the base.)
2. With the pin free to move in its fit, weld the centerline support plate to its restraining block. (Refer to MLI 1603 for details). Bend and tack weld the pin strap to the support block.
3. With the base forward guide block resting on its restraining plate and centered in the centerline support block (see Fig. 47), install temporary shims or keys to maintain the centerline support guide in position and weld the guide block to the restraining plate.
4. To minimize movement of the centerline support guide or during welding an alternating side stitch weld should be used.
5. Fit and install the permanent keys to accommodate any movement that may have occurred during welding. Clearance is (typically) 0.001 to 0.003 inches on each side. Bend the key locking strips to secure the assembly.

Load Compartment

1. After alignment, install load compartment on shims supplied. Install compartment seals between load compartment and generator, then between load compartment and the turbine.
2. Install all necessary conduit and piping/tubing which penetrates through the compartment walls.
3. Seal around penetrations. Install compartment flashing and conduct light check to locate compartment leaks. Seal leaks with high temperature sealant.
4. Install CO2 dampers and gravity dampers on the load compartment roof. Install ventilation ducting above these. Make CO2 fire protection tubing connection to CO2 dampers.
5. After Lube oil flushing, install the load compartment walkways.
6. Install load compartment ventilation fans. (See Figs. 48 and 49).

Installing Turbine Vent Fans

1. Remove turbine compartment temporary ventilation panels.
2. Install assembled turbine vent fans and ducting on turbine roof.
3. Check CO2 and gravity actuated damper orientations.
4. Check fan inlet diffuser for rubbing against fan. If this exists loosen and reposition the diffuser central to fan blades.
5. Connect CO2 fire protection tubing lines to CO2 dampers.

Installing All Off Base Modules

Place the following on foundations; Cooling Water skid, Water Wash Skid, Water Injection Skid, Gas Fuel Skid, Lube Oil Demister, GAC, PEECC and Exhaust Frame Blowers. Refer to Mechanical outline drawing MLI 0306 for skid elevations.

Installing Cooling Water Piping

1. Assemble cooling water piping (MLI 0914) on the accessory and turbine base. (See Figs. 50, 51, 52, and 53).
2. Inspect assembled piping flanges and check for pipe stress, then correct.
3. Install all orifices and check orientation.
4. Install temperature regulating valves and check orientation. (Refer to EM 5080).



Figure 45. Cutting Down Anchor Bolts

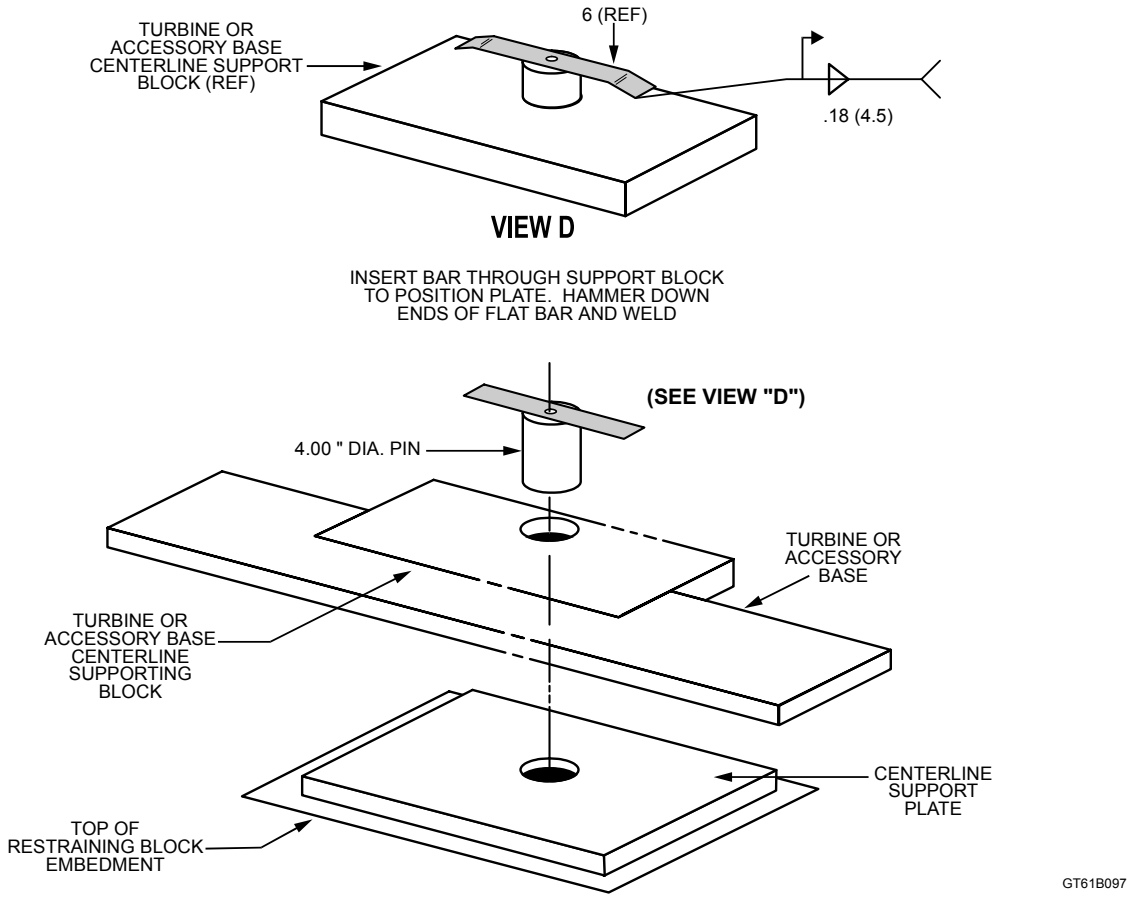


Figure 46. Turbine and Accessory Base Support Plate

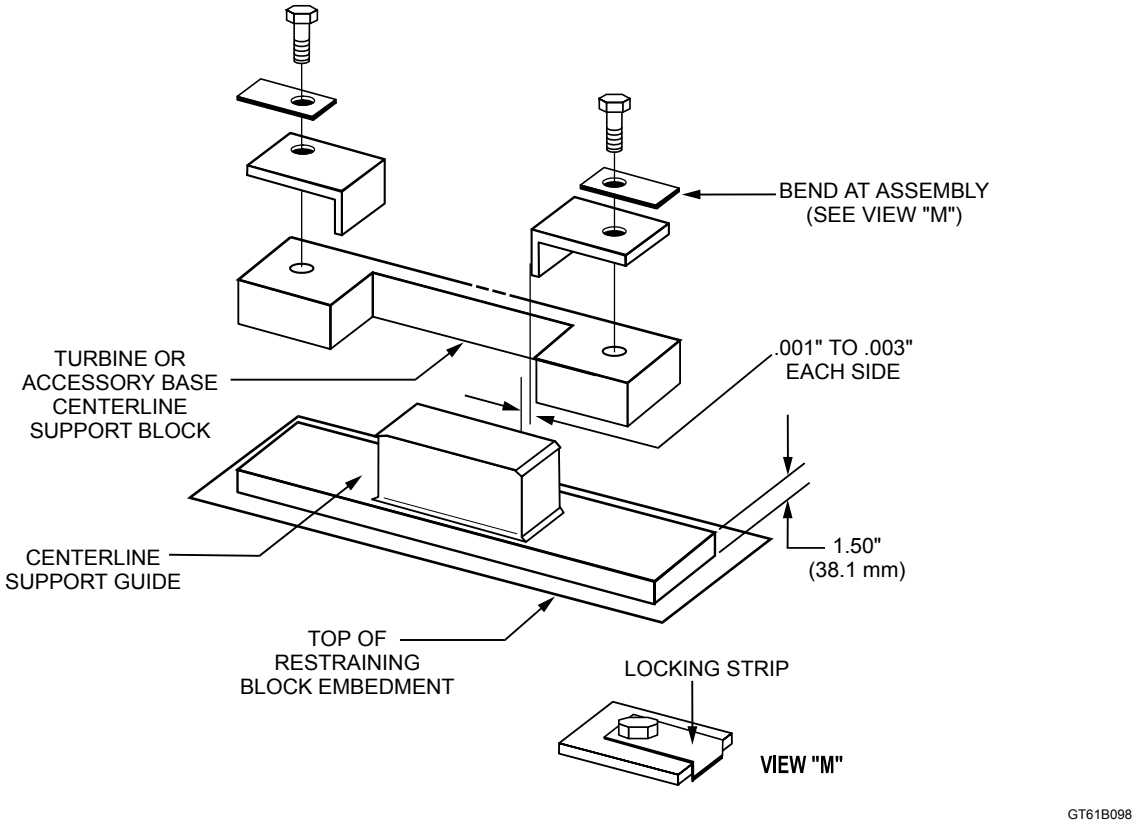


Figure 47. Turbine Accessory Base Support Guide

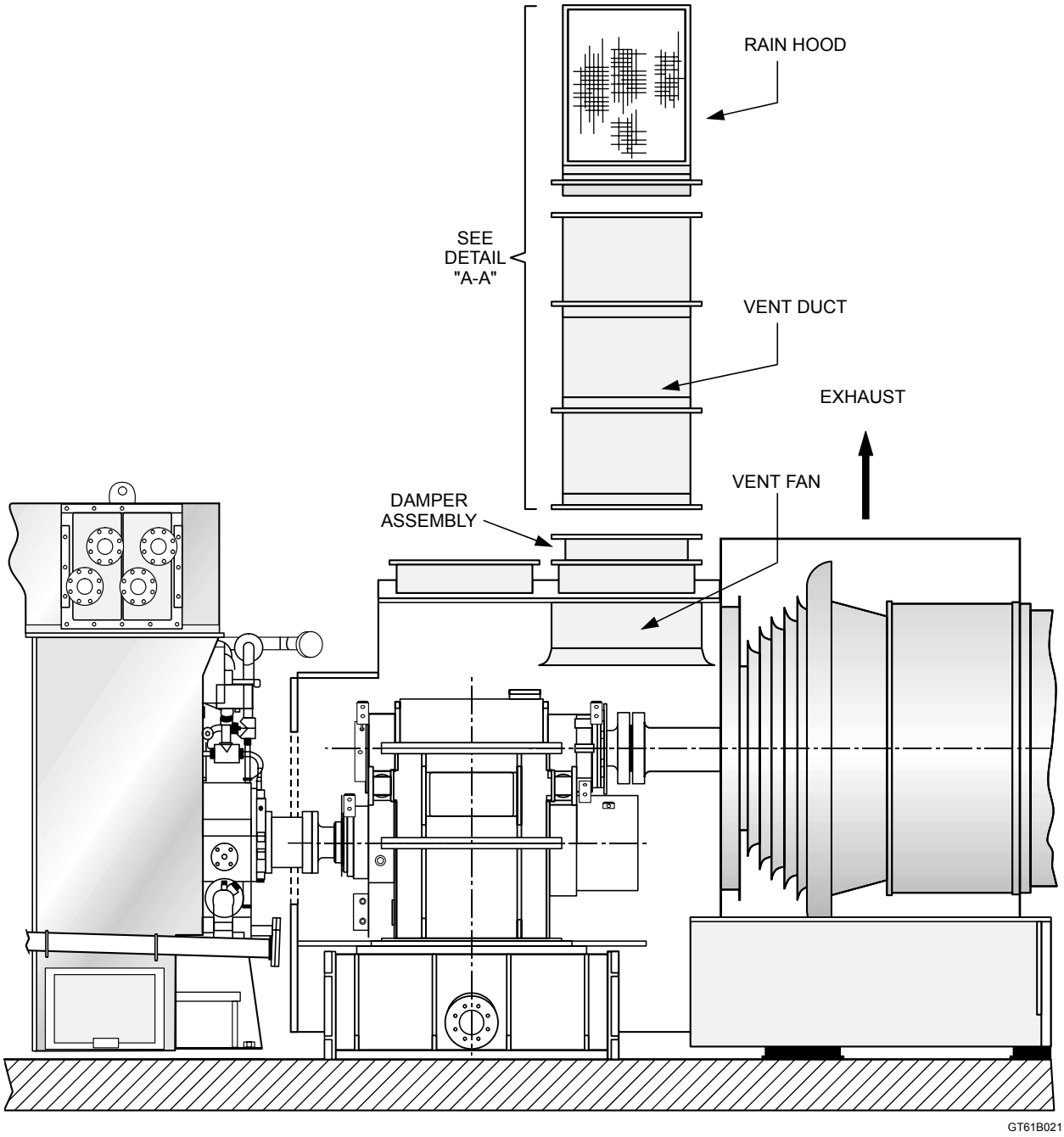


Figure 48. Air Inlet and Vent Duct Assembly



Figure 49. Assembling Load Compartment Ventilation Ductwork



Figure 50. Assembling Lube Oil Cooling Water Heads & Piping



Figure 51. Assembling Cooling Water Piping



Figure 52. Assembling Cooling Water Piping

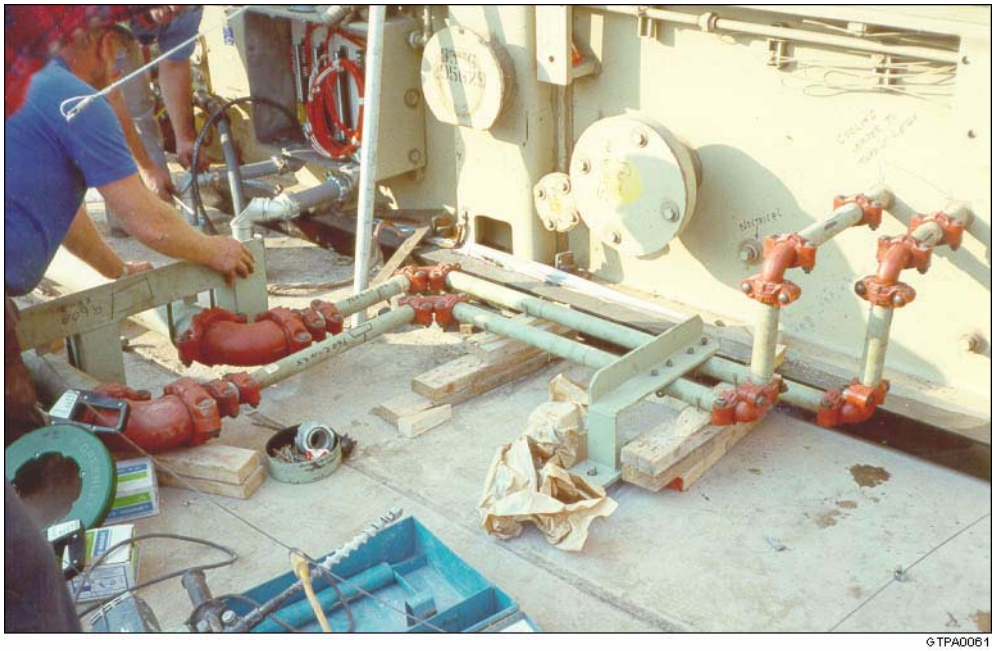


Figure 53. Assembling Cooling Water Piping (to cool turbine aft support legs)

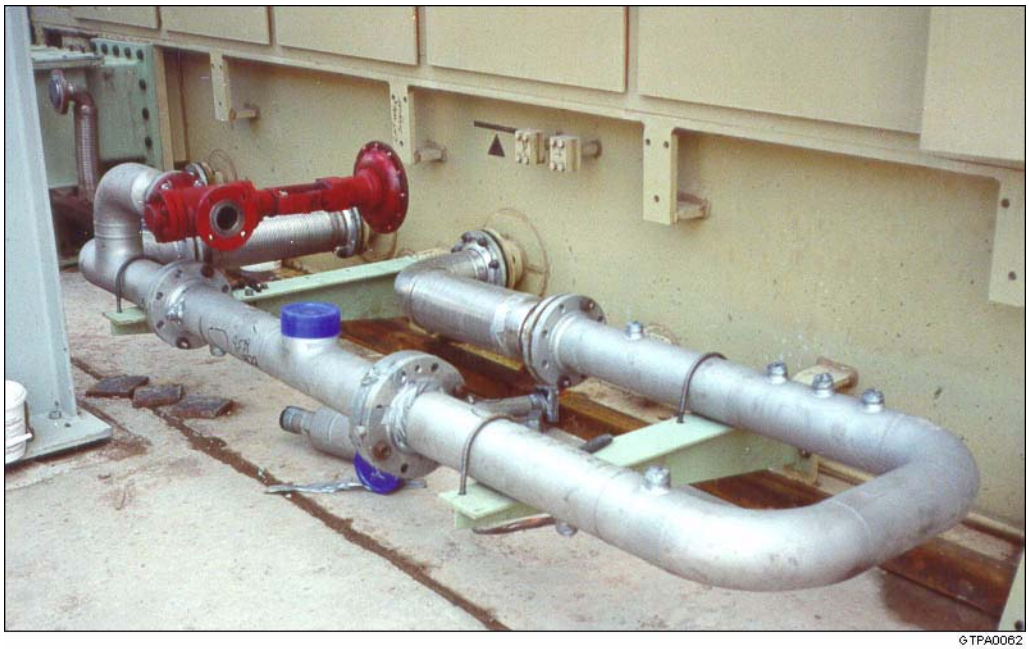


Figure 54. Assembling Lube Oil Heads at Left Side of Accessory Base

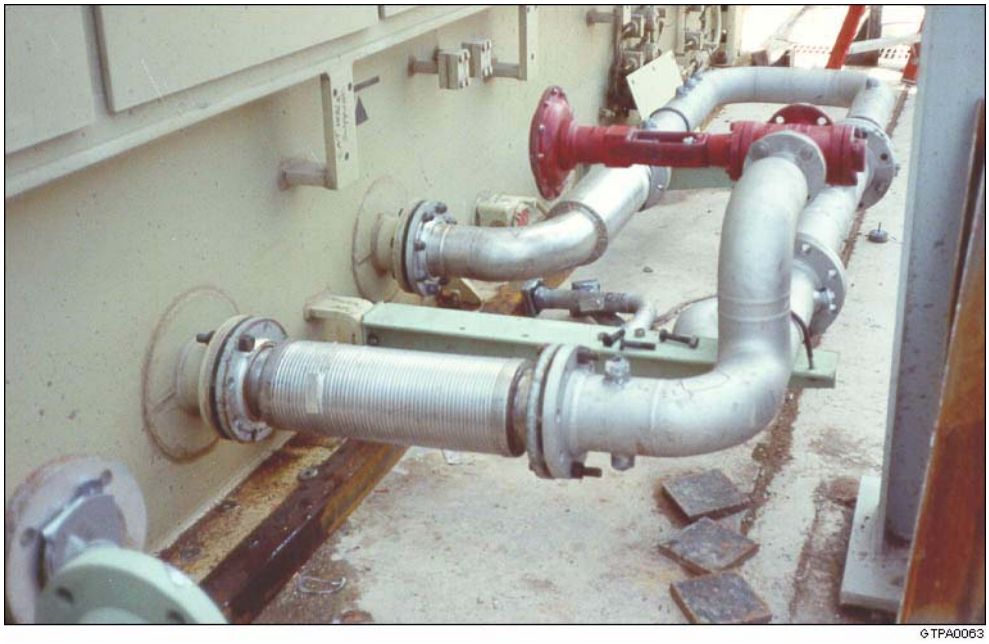


Figure 55. Assembling Lube Oil Heads at Left Side of Accessory Base

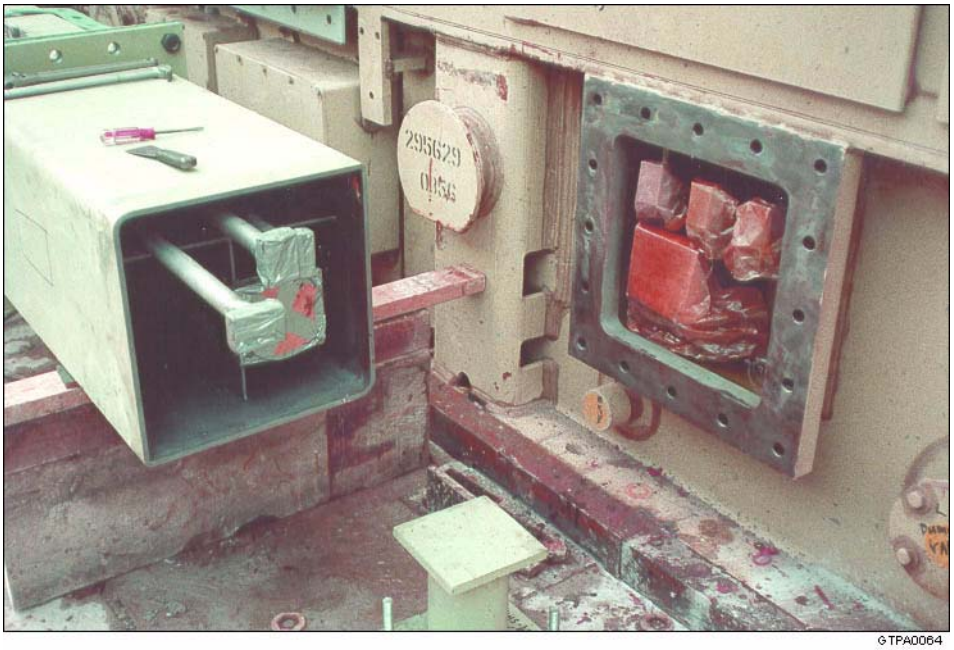


Figure 56. Assembling Lube Oil Heads at Left Side of Accessory and Turbine Base

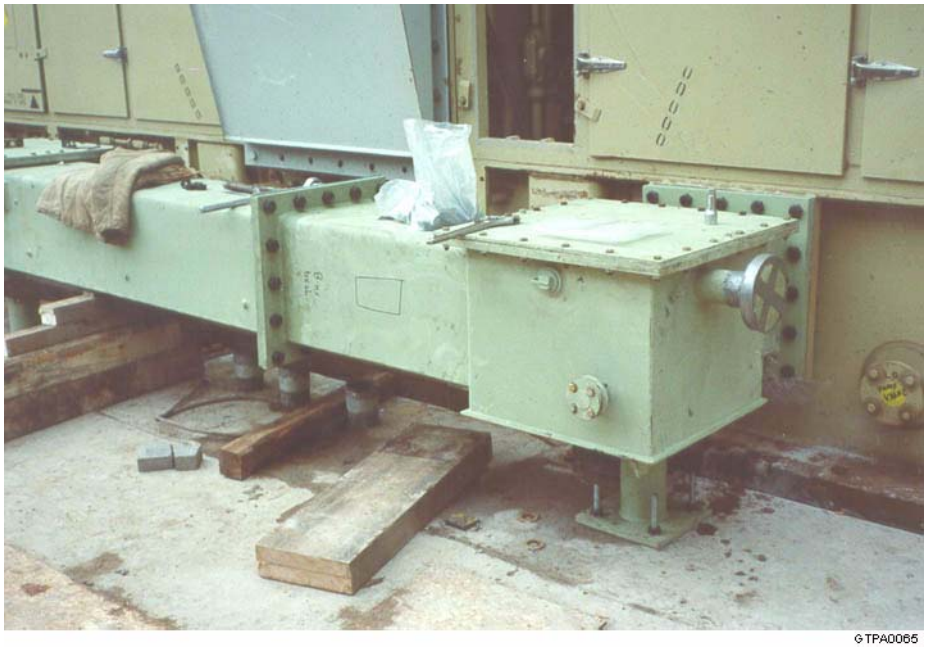


Figure 57. Assembling Lube Oil Heads at Left Side of Accessory and Turbine Base

5. Use temporary gaskets where piping will be disassembled for flushing the system.
6. Flush the cooling water system in sections.
7. Remove dirt contained in customer piping between on base equipment and the off base Cooling Water Skid first.
8. Bypass all equipment one by one and introduce each piece of equipment into the flush by restoring system piping, as follows:
   a. Disconnect the inlet and outlet Purchaser Connection piping to the finned fan heat exchanger, generator and turbine.
   b. Fabricate piping spools so that the water flow bypasses all the above equipment.
   c. Remove both temperature regulating valves and install “T” spool pipes in their place.
   d. Pump water through the system (with spools installed), inspect and clean the “Y” strainer on the cooling water skid pump suction line until it is clean.
   e. Next remove the spool pipe bypassing the finned fan heat exchange and flush though it until clean water is obtained.
   f. Again remove the generator bypass spool pipe and flush through the heat exchanger and generator.

g. Then remove the turbine bypass spool pipe and flush the whole system until it is clean.

h. Reinstall the temperate regulating valves and continue flushing until strainer is clean and then finish flush.

i. Add corrosion inhibitor to the cooling water system as specified in GEI-41004.

9. Install temperature regulating valve capillary tubes into lube oil header and atomizing air piping.

Installing Lube Oil Piping

1. After alignment install Lube oil piping between Accessory base, Turbine, Load Gear and Generator. (See Figs. 54, 55, 56 and 57).
2. Fit all piping to equipment and check flanges for pipe stress, then correct if necessary.
3. Use temporary gaskets while fitting piping where it will be removed and replaced with flushing piping for the lube oil flush.
4. Correct fitting of this piping before the lube oil flush will minimize the time to replace permanent piping after the flush, preventing contamination of the clean lube oil system.
5. Install the lube oil flush piping (A0125) and conduct the flush. Refer to Dwg. 354A1481 for flushing procedures.
6. Finish the flush when the installed “Y” strainers and system are clean meeting the procedure’s criteria.
7. Restore the permanent lube oil piping with permanent gaskets. All piping being restored should be cleaned and inspected before installation. (Refer to EM 2420 – Maintenance Flexitallic Gaskets).

Remaining Piping and Walkways

1. Install remaining Cooling and Sealing air piping, including compressor bleed lines and cooling and sealing line to No. 2 bearing.
2. Install all remaining interconnecting piping including; liquid fuel lines, gas fuel lines, false start drain lines, and Atomizing Air lines.
3. Install all remaining piping to Purchaser Connection points including all water wash low point drain lines, exhaust frame blower lines, water injection and water wash lines, telltale leakoff line, and liquid fuel line.
    a. Both the water wash and water injection lines connecting to the Turbine should be disconnected and flushed by pumping water through the respective skids. The piping can then be reconnected to the Turbine.
    b. Also the Exhaust frame blower line to the Turbine should be disconnected and the blowers turned on to flush out any dirt that might block orifices in the Turbine casing.
4. Liquid fuel lines should be flushed up to the main stop valve in the Accessory base to remove any dirt up to that point. On base liquid fuel lines should be disconnected after the flow divider and before the fuel nozzle check valves and blown down with compressed air before reconnecting.
5. Unit walkways should be installed except for load compartment walkways which can be installed after the lube oil flush.

Setting Lube Oil and Hydraulic Pressures

After the lube oil flush is complete and the lube oil system is restored, set the lube oil header pressure by adjusting VPR2-1 while the Auxiliary pump is running. This pressure can be re-adjusted if necessary during operation when the gear driven Lube oil pump is in operation. Reference MLI 0416.

Next set the Hydraulic pressure for the Auxiliary Hydraulic pump, as follows:

1. Adjust both VR22-1 and VPR3-2 together and bring the Hydraulic pressure on the gauge panel above the pressure setting of VR22-1 in the device summary.
2. Adjust VR22-1 and bring the hydraulic system pressure back to the device summary setting. This is set now for operation.
3. Adjust VPR3-2 now bringing the pressure down to the setting in the device summary. This is now set for operation.

To set the VPR3-1 on the gear driven hydraulic pump and the associated VR21-1, follow the same procedure while the machine is in operation.

To set the DC Emergency lube oil pump discharge pressure, set the speed at the recommended nameplate operating speed. Monitor the amps used by the motor while not exceeding the nameplate rating. Check the pump discharge pressure and notify Engineering if it differs from that given in MLI 0416.

ASSEMBLY OF COUPLINGS

Assembly of Load Coupling

Refer to EM 5316 (MS61B–Lucas Airospace Load Coupling Inst.)

NOTE: During assembly of the load coupling, it is important that the vendor drawing notes be closely followed to insure proper coupling assembly. The following procedure is typical for the coupling currently being utilized with the MS6001B.

Lucas/Bendix Load Coupling

This coupling consists of four pieces. Two adapter plates and two center pieces (Fig. 58). The adapter plates each have a typical interference rabbet fit of 0.002 to 0.004 inches, and are bolted to the load gear hub and the turbine rotor hub, respectively.

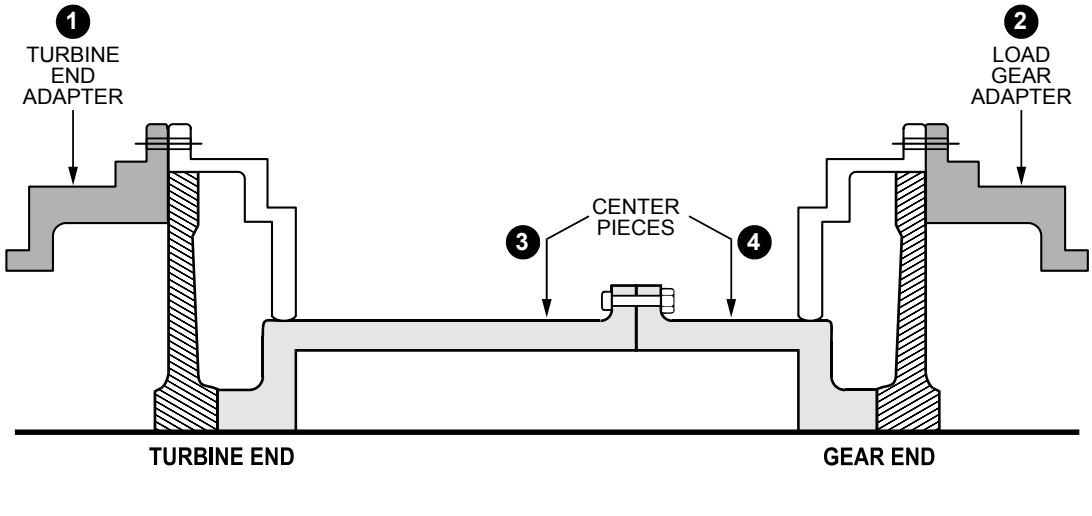


Figure 58. Bendix Load Coupling

The general procedure for assembly of this coupling is first to install the two adapter plates on the respective turbine hub and load gear hub. Next, the longer section of the coupling is assembled to the turbine coupling hub adapter plate. Then, the shorter section of the coupling is assembled to the (already assembled) longer section. Finally, the coupling is made up to the load gear coupling hub adapter plate. For more specific details on this assembly procedure, refer to the vendor drawings and notes.

1. Measure and record the length of each of the adapter plate coupling bolts. Use of a modified dial caliper has proven beneficial in obtaining these measurements. (See Figs. 59, 60 and 61).
2. Weigh and record (in grams) each nut and bolt assembly in order to pair up equal weight assemblies, 180 degrees apart, to obtain optimum balance.
3. Pack both of the adapter plates in dry ice (overnight if possible) to enhance the assembly of the interference fits at the turbine and the load gear coupling hubs. When the adapters are sufficiently shrunk, they can be assembled to their respective hubs. (See Fig. 62).
4. Apply a light coat of Fel-Pro C-102, or equivalent, to the bolt threads and assemble in the adapter plate.
5. Initially torque all the bolts to 350-400 ft-lbs. (Refer to MLI 0411). Note that a hydraulic torque wrench will be needed to torque the bolts because of a lack of space in the area, and the turbine will have to be rotated at least 180 degrees.

NOTE: Minimize turbine rotation as much as possible since lubrication is not provided to the bearings.

6. Stretch the bolts between 0.0095 and 0.0115 inch. (Refer to MLI 0411). About 1600 to 1640 ft-lbs of torque will be required. The turbine will need to be rotated 180 degrees again.
7. The longer section of the coupling will now be assembled to the turbine coupling hub adapter flange. Matchmarks on these pieces must be in line.
8. Apply a thin coat of Fel-Pro to the bolt threads and, while supporting the free end of the coupling, draw up the coupling section to the turbine adapter plate, ensuring that the rabbet fit is properly engaged. Bolts are to be torqued to 60 ft-lbs. Support the free end of the coupling throughout this procedure. The turbine will again have to be rotated 180 degrees.

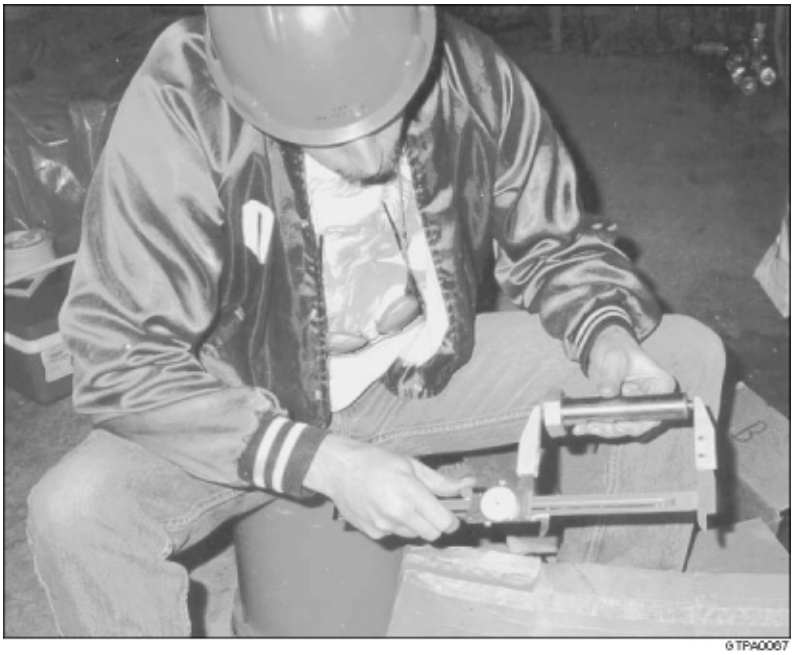


Figure 59. Measuring Coupling Bolts with Modified Dial Caliper

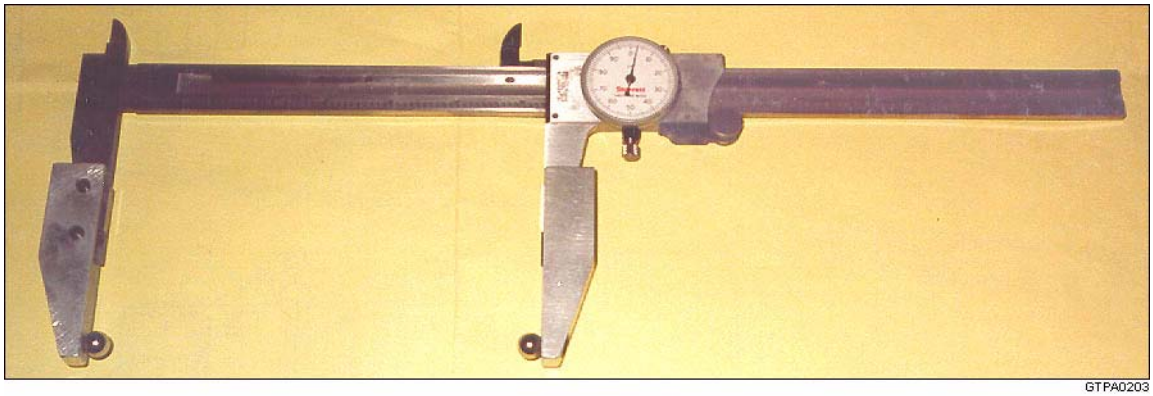


Figure 60. Modified Dial Caliper for Measuring Coupling Bolts

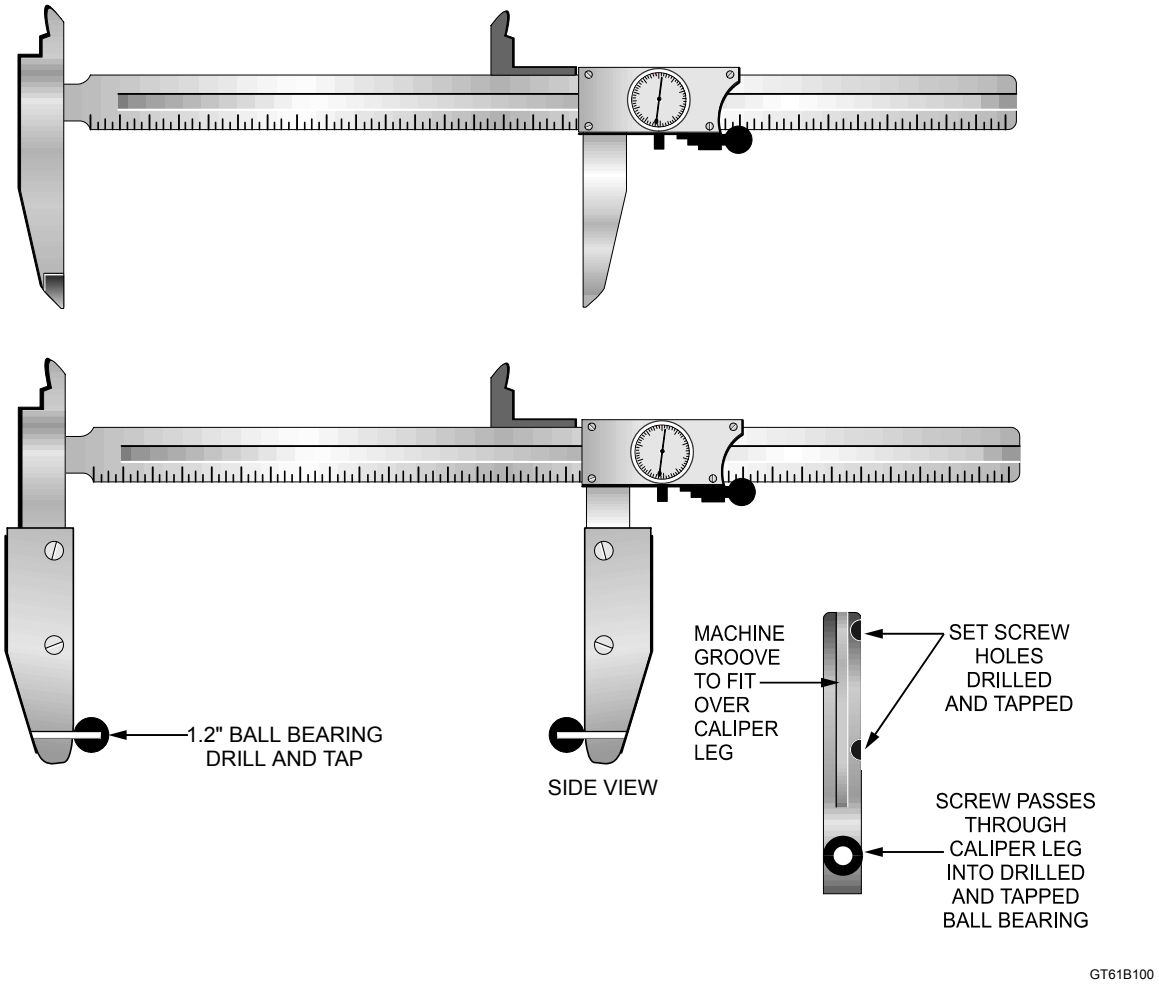


Figure 61. Dial Caliper (14 Inch – Starrett type)



Figure 62. Coupling Adapter Plate Packed in Dry Ice

9. Before the short section of the coupling can be assembled to the longer section, it is necessary that the load gear high-speed shaft be floated aft about 1 inch to allow room for the assembly of the short section, i.e., clear the rabbet fit. Rotate the gear shaft slightly while moving the gear rotor in the aft axial direction.

10. Install the short section of the coupling and bolt to the already assembled longer section. Matchmarks must be in line. Bolts should be torqued to 100 ft-lbs. Maintain support for the free end of the coupling.

NOTE: See vendor drawings for details on assembly of coupling shims.

11. Make up the short coupling section to the load gear coupling hub adapter flange. Move the load gear rotor forward toward the turbine while slowly engaging the rabbet. Matchmarks must be in line. Both the turbine and the load gear will have to be rotated while aligning the bolt holes and while engaging the rabbet. Bolts should be lubricated with Fel-Pro and torqued to 60 ft-lbs.

Turn Load Coupling

NOTE: This coupling is no longer supplied and has been removed from service at most installations.

Assembly of Generator Coupling

1. Measure and record the length of each of the coupling bolts with a modified dial caliper (Figs. 59, 60 and 61). Each of the coupling bolts has to be stretched by means of an hydraulic torque wrench. These bolts will be found in the generator loose parts MSD.
2. Determine and record the weight in grams of each nut and bolt assembly in order to pair up the bolts as close as possible. (Pairs are assembled 180 degrees apart in the coupling for balance).
3. Lubricate the coupling bolts with a thin coat of Molycoat or Fel-Pro C5A or similar thread lubricant.
4. Use the bolts to engage the generator rotor coupling flange rabbet into the mating gear flange and pull the coupling flanges together.

NOTE: There is an interference fit in the rabbet of about 0.002 inches, but the coupling will make up without much difficulty.

5. Ensure that the rabbet is engaging properly before tightening the coupling bolts.
6. Use an hydraulic torque wrench to stretch bolts from 0.0095 to 0.0115 (about 1600 to 1640 ft-lbs of torque will be required. (Refer to MLI 0411).
7. Remove the load gear centering ring which cannot be left on the shaft. (See Figs. 19, 20, and 63 Load Gear Centering Ring).

Assembly of Accessory Coupling

Currently, two different manufacturers supply accessory couplings; Zurn and Lucas Aerospace (Bendix). Both are a three-piece coupling and are fairly easy to assemble; however, the turbine end adapter plate requires a special torque wrench adapter which is shown in Fig. 64. (Refer to MLI 0411 for detailed instructions on assembly).

1. With one person supporting the accessory gear adapter plate from atop the accessory gear, a second person should install the coupling bolts.
2. Install the turbine rotor adapter plate.
3. Use the special torque wrench adapter to torque the turbine rotor adapter plate bolts. (See Fig. 64).
4. Assemble the accessory gear end of the coupling distance piece.
5. Bolt a strap across the turbine end bolting flange in order to draw the coupling diaphragm up to bolt that end.

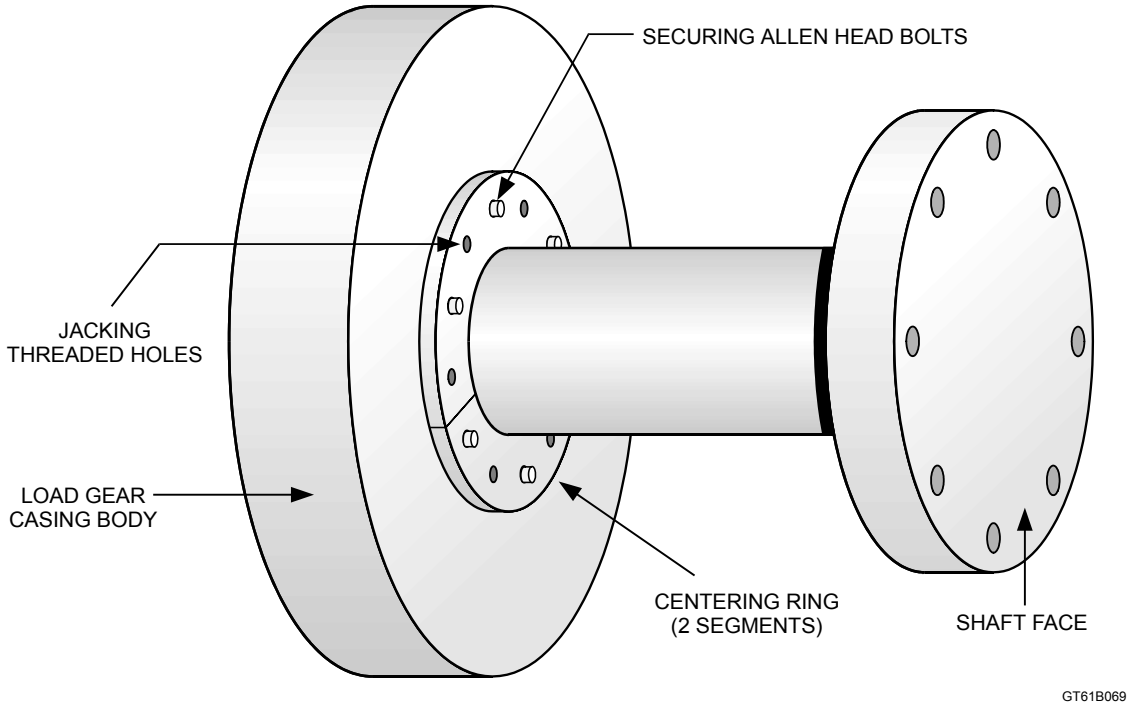
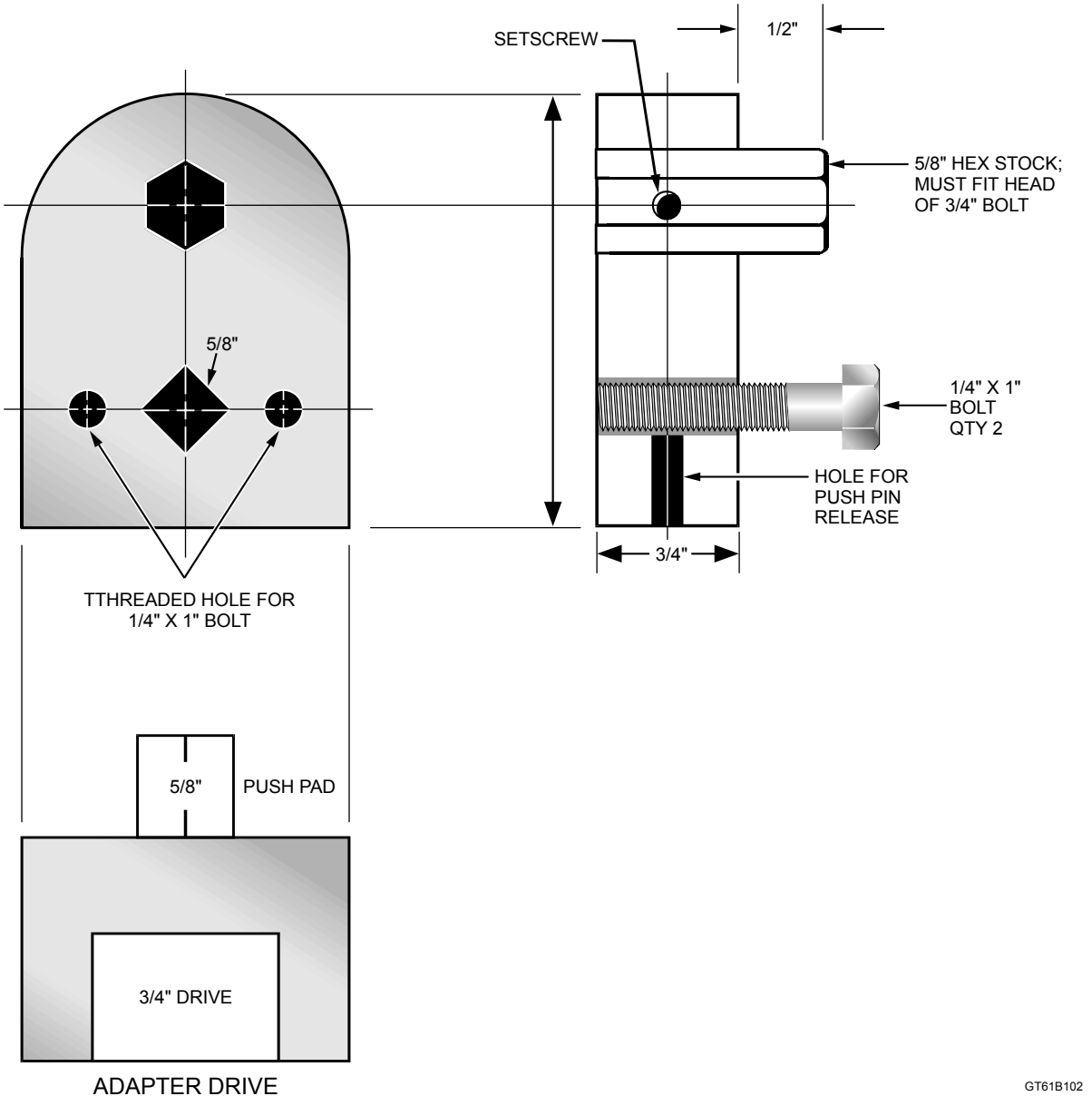


Figure 63. Quill Shaft Flange – Load Gear



Note: This adapter must be fabricated.

Figure 64. Torque Wrench Adapter (for Bendix Accessory Coupling Adapter Plate, Turbine Rotor End)